INCH-POUND

MIL-M-28787/364
23 July 1990

MILITARY SPECIFICATION

MODULES, DIGITAL, STANDARD ELECTRONIC

REGISTER MODULE, KEY CODE DGJ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic register module. This module is class I as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards and handbook. The following specification, standards and handbook form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

HANDBOOK

MILITARY

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a failure rate predicted in accordance with MIL-STD-1389. The predicted failure rate shall not exceed 0.6355 failures per million operating hours when predicted in accordance with DOD-HDBK-217E.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be DGJ.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 120 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat-dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
A	Isolation,									
	initial	II	S	10	INF	10	INF	10	INF	MOhm
	end-of-life	II	Q	1	INF	1	INF	1	INF	MOhm
B1-B4	Continuity	III	X	0.0	1.0	0.0	1.0	0.0	1.0	Ohm
B5-B8	Continuity	III	Q	0.0	0.1	0.0	0.1	0.0	0.1	Ohm
C	Vcc stress	IV	Q	--------Module shall pass all remaining tests----------						
D1-D28	Ibd	V	S	0	100	0	100	0	100	uA
D29-D60	Ibd	V	S	0	120	0	120	0	120	uA
E1-E60	Vic	VI	S	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	V
F1-F32	Ios	VII	S	-225.0	-40.0	-225.0	-40.0	-225.0	-40.0	mA
F33-F64	Ios	VII	S	-225.1	-40.1	-225.1	-40.1	-225.1	-40.1	mA
F65	Ios	VII	S	-100.4	-15.0	-100.4	-15.0	-100.4	-15.0	mA
G1-G28	Iih	VIII	X	0	20	0	20	0	20	uA
G29-G60	Iih	VIII	X	-20	40	-20	40	-20	40	uA
H1-H28	Iil	IX	X	-400	-30	-400	-30	-400	-30	uA
H29-H60	Iil	IX	X	-420	-80	-420	-80	-420	-80	uA
J1-J64	Voh	X	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J65	Voh	X	X	2.5	4.5	2.5	4.5	2.5	4.5	V
J66	Voh	X	X	2.4	2.78	2.4	2.78	2.4	2.78	V
K1-K65	Vol	XI	X	0.0	0.4	0 0	0.4	0.0	0.4	V
L1-L32	Iozh	XII	X	-20	20	-20	20	-20	20	uA
L33-L64	Iozh	XII	X	-20	40	-20	40	-20	40	uA
M1-M32	Iozl	XIII	X	-20	20	-20	20	-20	20	uA
M33-M64	Iozl	XIII	X	-420	-80	-420	-80	-420	-80	uA
N	Icc	XIV	X	30	585	30	585	30	585	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
P1,P2	Functional	XV	X	See table XV						
Q1,Q2	Dynamic	XVI	S	See table XVI						
Q3,Q4	Dynamic	XVI	Q	See table XVI						
R	PSFC	XVIII	S	4.0	7.4	4.0	7.4	4.0	7.4	uF
S	PSTR	XIX	Q	0	400	0	400	0	400	mV
T1-T66	Output crosstalk (high level)	XX	Q	0	500	0	500	0	500	mV
U1-U32	Output crosstalk (low level)	XXI	Q	0	900	0	900	0	900	mV
U33-U65	Output crosstalk (low level)	XXI	Q	0	500	0	500	0	500	mV
V1-V32	Crosstalk duration (low level)	XXII	Q	0	70	0	70	0	70	nS
W1-W27	Cin	XXIII	Q	2	30	2	30	2	30	pF
W28	Cin	XXIII	Q	2	50	2	50	2	50	pF
X1-X32	Cout	XXIV	Q	2	40	2	40	2	40	pF
X33-X64	Cout	XXIV	Q	2	45	2	45	2	45	pF
Y	Threshold	XXV	X	See table XXV						

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
Z1-Z32	Disable time, tPLZ	XXVI	Q	4	45	4	59	4	59	nS
Z33-Z64	Disable time, tPHZ	XXVI	Q	4	50	4	65	4	65	nS
Z65-Z96	Disable time, tPLZ	XXVI	Q	6	61	6	89	6	89	nS
Z97-Z128	Disable time, tPHZ	XXVI	Q	6	66	6	95	6	95	nS
Z129-Z160	Disable time, tPLZ	XXVI	Q	6	61	6	89	6	89	nS
Z161-Z192	Disable time, tPHZ	XXVI	Q	6	66	6	95	6	95	nS
Z193-Z224	Disable time, tPLZ	XXVI	Q	4	45	4	59	4	59	nS
Z225-Z256	Disable time, tPHZ	XXVI	Q	4	50	4	65	4	65	nS
Z257-Z288	Disable time, tPLZ	XXVI	Q	6	61	6	89	6	89	nS
Z289-Z320	Disable time, tPHZ	XXVI	Q	6	66	6	95	6	95	nS
Z321-Z352	Disable time, tPLZ	XXVI	Q	6	61	6	89	6	89	nS
Z353-Z384	Disable time, tPHZ	XXVI	Q	6	66	6	95	6	95	nS
AA1-AA32	Noise margin	XXVII	Q	0	900	0	900	0	900	mV

- See footnotes on next page.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/364-1. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. The module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.0 ± 0.5 V dc, except the output short circuit current limits apply only for a power supply voltage of 5.5 V dc ± 1 percent.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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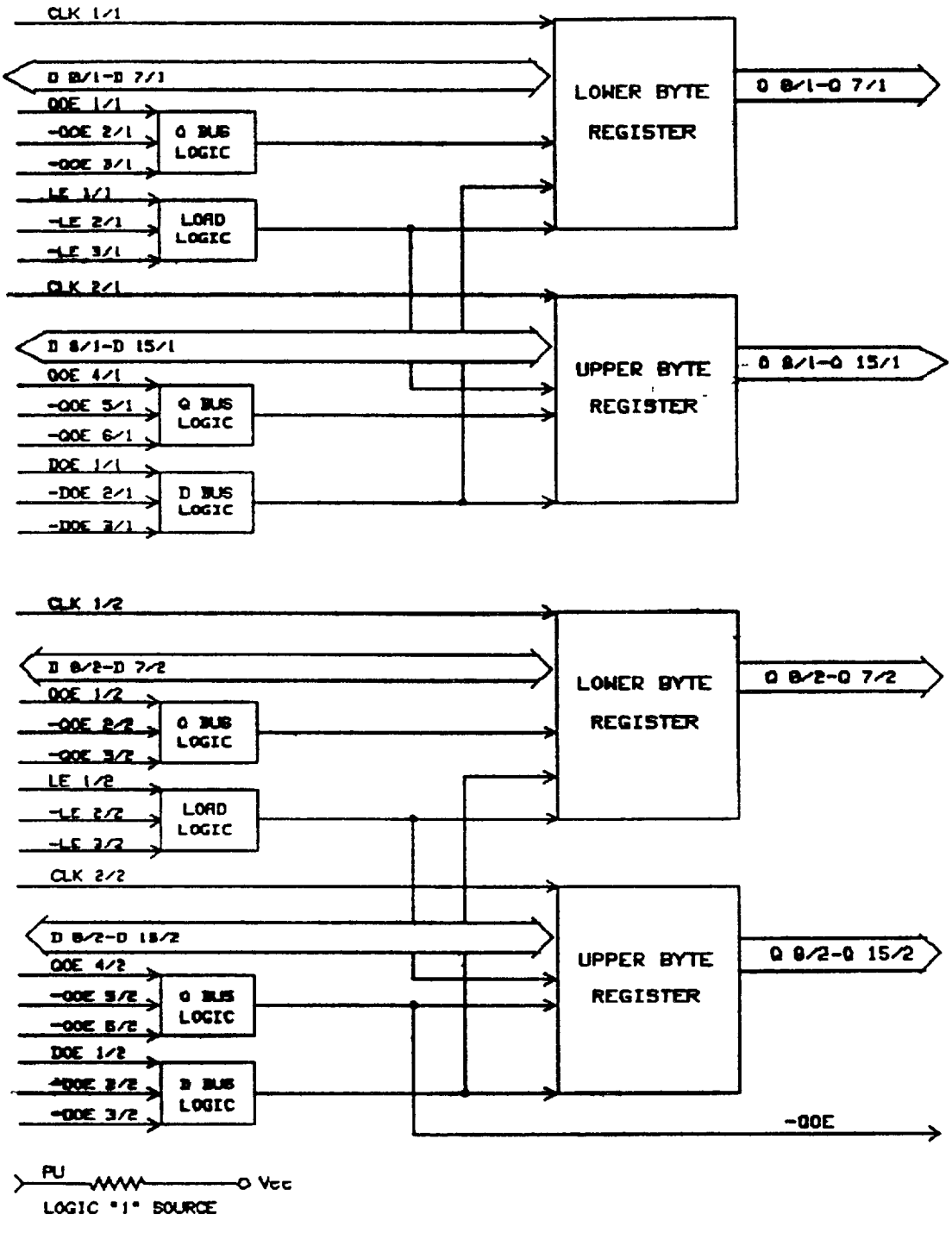


FIGURE 1. DGJ functional block diagram.

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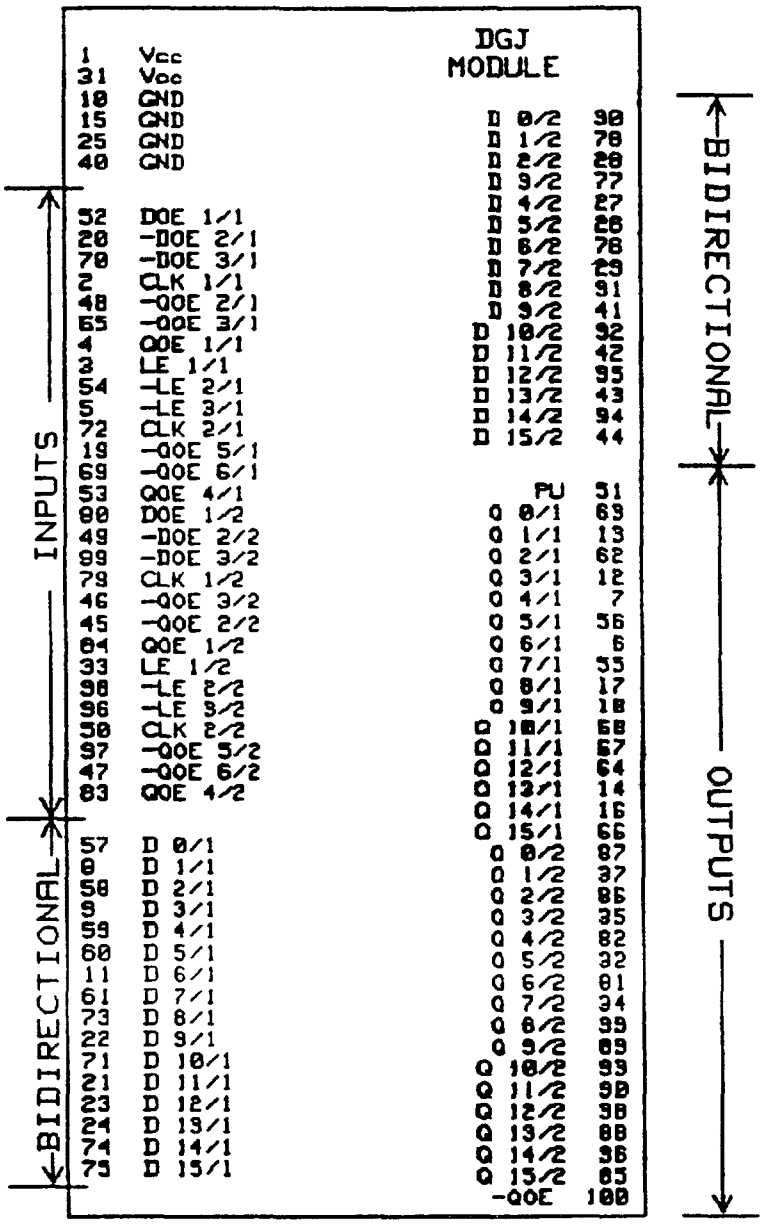


FIGURE 2. DGJ pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through AA shall be performed in any order with the following exceptions. Test conditions A, C, D, E and F are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or F are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E and F where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

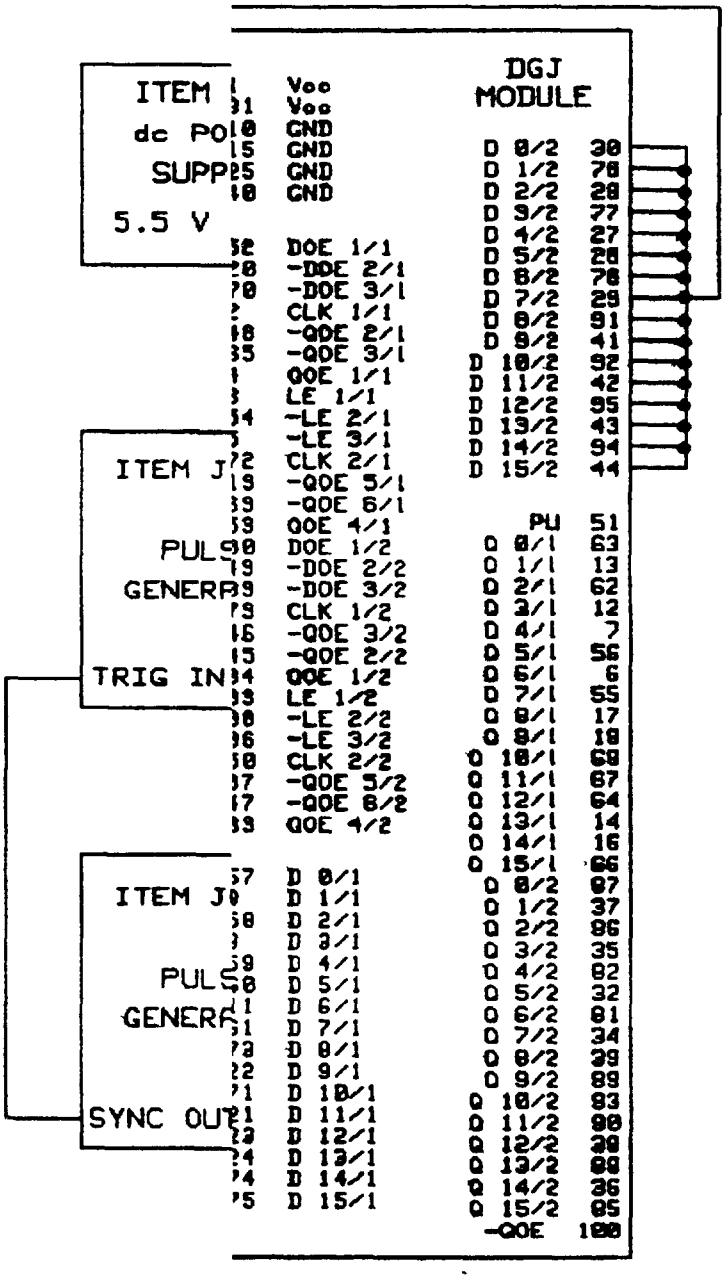
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through AA in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, G through V, Y, Z and AA shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated as shown on figures 3 and 4.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figures 5 and 6.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 7. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition R, shall be performed using the initial limits at 25 ± 5°C ambient.

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NOTES:

1. Load output voltage test.

2. R1 = R2 =

FIGURE 3. and life test circuit.

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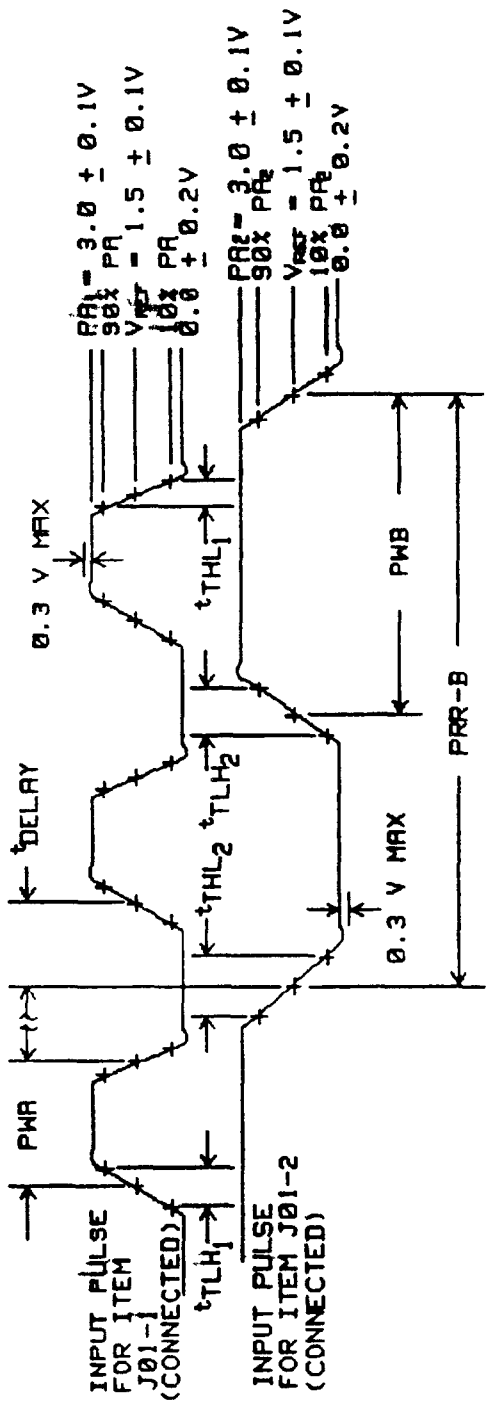


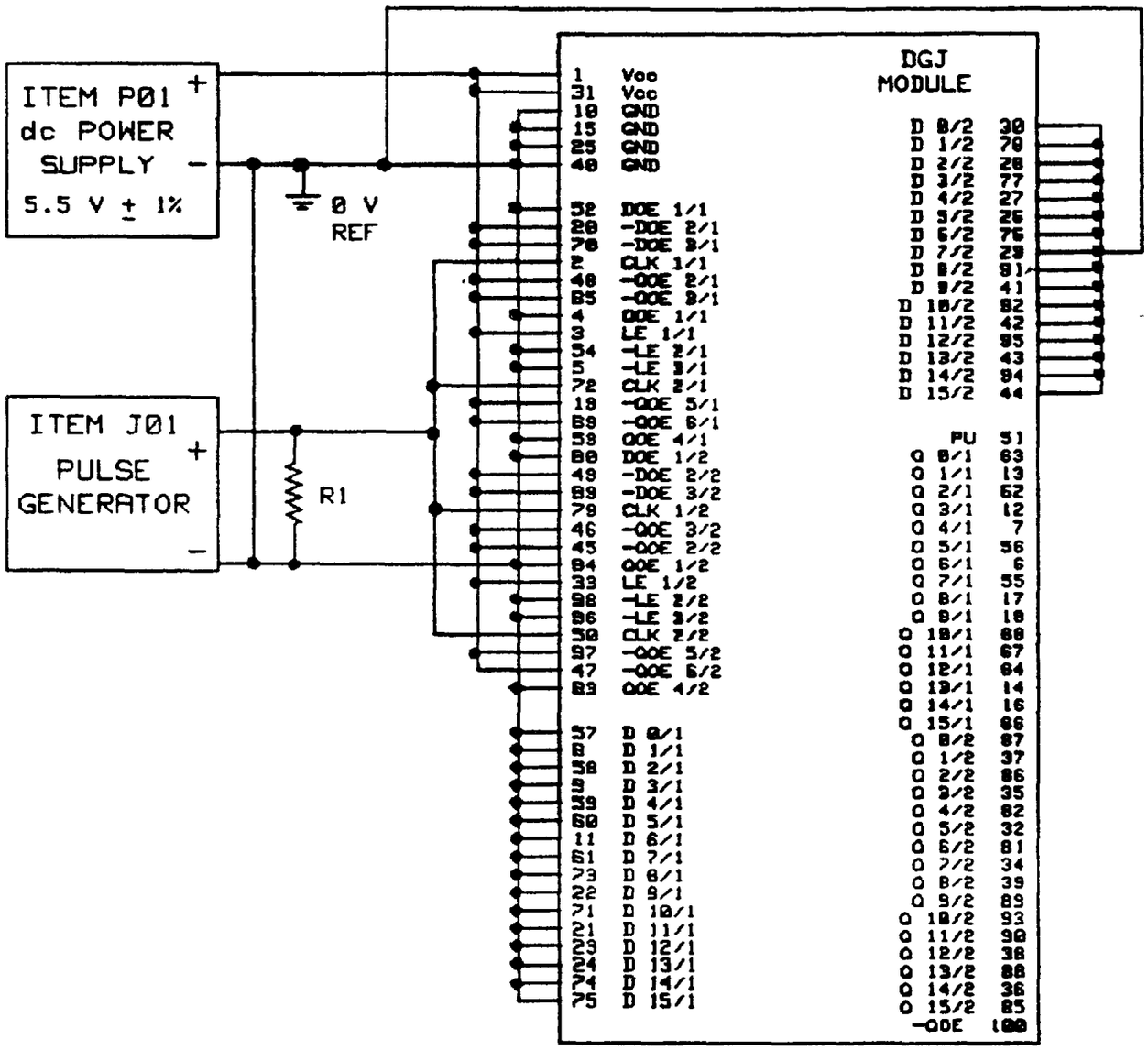
FIGURE 4. Operating temperature and life test waveforms.

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NOTES:

1. PWA = 250 ± 25 nS.
2. tTLH1 = tTHL1 = 50 ± 30 nS.
3. tDELAY = 150 ± 50 nS.
4. PWB = 500 ± 50 nS.
5. tTLH2 = tTHL2 = 50 ± 30 nS.
6. PRR-B = 1000 ± 100 nS.
7. All voltages are referenced to ground.

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NOTES:

1. Load outputs as specified in the output voltage test.
2. R1 = 51 Ohms ± 5 percent, 1 W.

FIGURE 5. DGJ component temperature test circuit.

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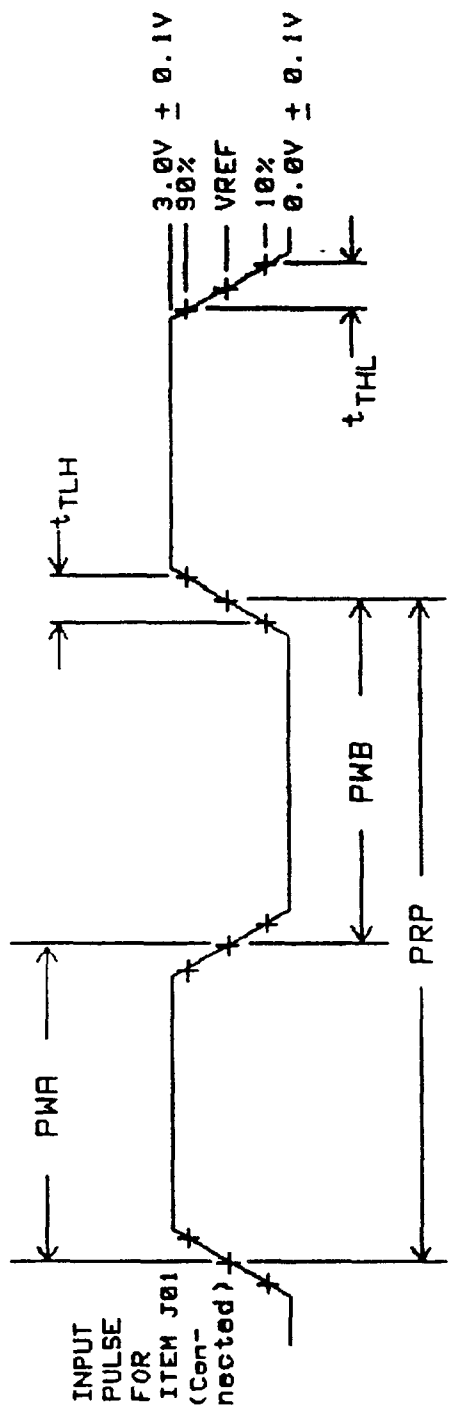


FIGURE 6. Component temperature test waveforms.

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NOTES:

1. PWA = 250 ± 25 nS.
2. PWB = 250 ± 25 nS.
3. PRP = 500 ± 50 nS.
4. tTLH = tTHL = 50 ± 10 nS.

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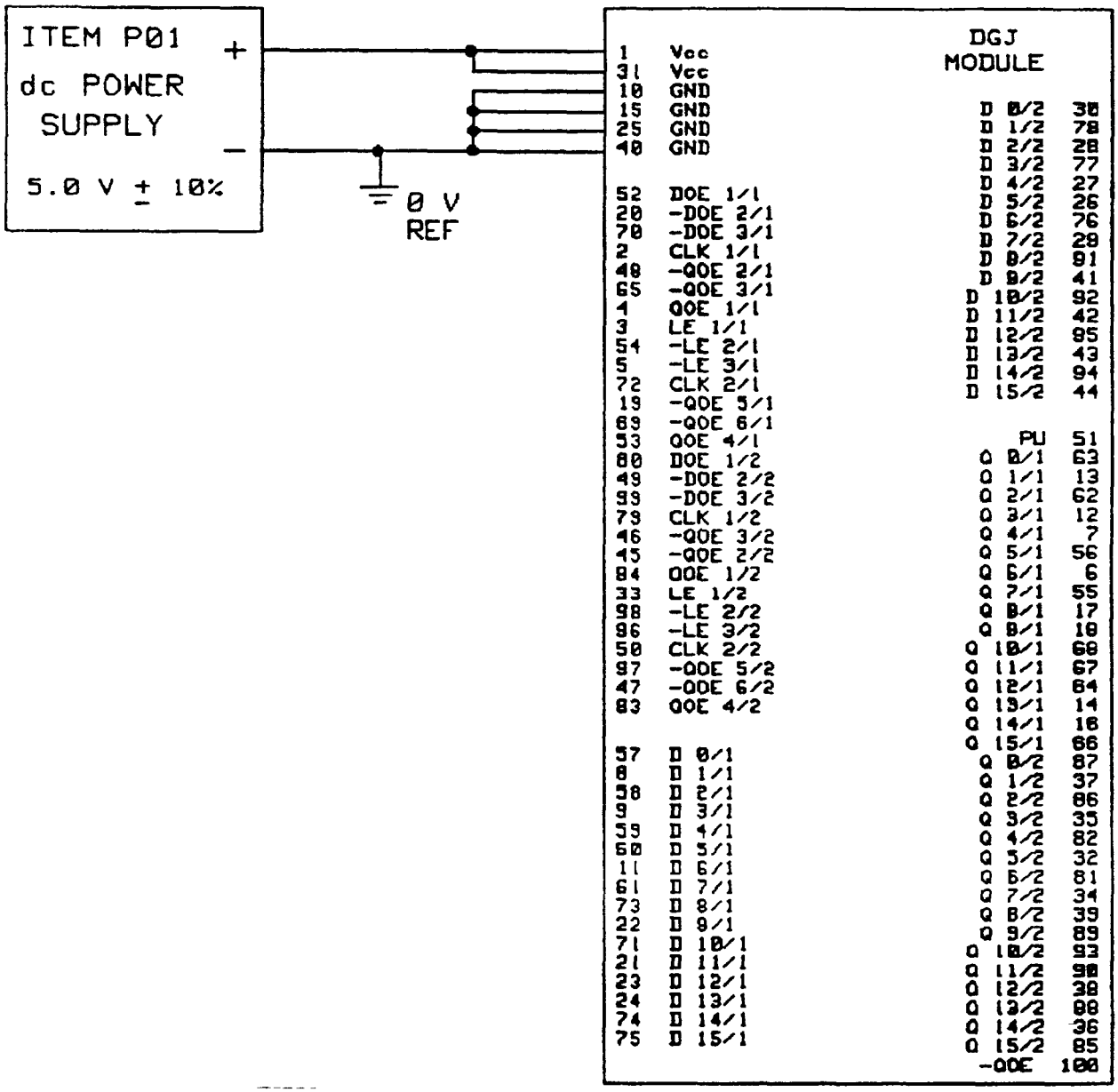


FIGURE 7. DGJ vibration test circuit.

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4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figures 3 and 4. Following the life test and while still at the maximum class temperature, electrical tests B, G through V, Y, Z and AA shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, G through V, Y, Z and AA shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Cout = output capacitance.
(c) Ibd = input breakdown current.
(d) Icc = power supply current.
(e) Iih = input high level current.
(f) Iil = input low level current.
(g) Iin = input forcing current.
(h) Ioh = output high level current.
(i) Iol = output low level current.
(j) Ios = output short circuit current.
(k) Iozh = output high level leakage current.
(l) Iozl = output low level leakage current.
(m) NA = not applicable.
(n) PSFC = power supply filter capacitance.
(o) PSTA = power supply transient amplitude.
(p) Vic = input clamp diode voltage.
(q) Vih = input high level voltage.
(r) Vil = input low level voltage.
(s) Vin = input forcing voltage.
(t) Voh = output high level voltage.
(u) Vol = output low level voltage.
(v) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 3.220 W (max); 1.724 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
DOE 1/1	-1.5	0.7	2.0	-0.4	20	100	30
DOE 1/2	-1.5	0.7	2.0	-0.4	20	100	30
-DOE 2/1	-1.5	0.7	2.0	-0.4	20	100	30
-DOE 2/2	-1.5	0.7	2.0	-0.4	20	100	30
-DOE 3/1	-1.5	0.7	2.0	-0.4	20	100	30
-DOE 3/2	-1.5	0.7	2.0	-0.4	20	100	30
CLK 1/1	-1.5	0.7	2.0	-0.4	20	100	30
CLK 1/2	-1.5	0.7	2.0	-0.4	20	100	30
CLK 2/1	-1.5	0.7	2.0	-0.4	20	100	30
CLK 2/2	-1.5	0.7	2.0	-0.4	20	100	30
QOE 1/1	-1.5	0.7	2.0	-0.4	20	100	50
QOE 1/2	-1.5	0.7	2.0	-0.4	20	100	30
-QOE 2/1	-1.5	0.7	2.0	-0.4	20	100	30
-QOE 2/2	-1.5	0.7	2.0	-0.4	20	100	30

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
-QOE 3/1	-1.5	0.7	2.0	-0.4	20	100	30
-QOE 3/2	-1.5	0.7	2.0	-0.4	20	100	30
QOE 4/1	-1.5	0.7	2.0	-0.4	20	100	30
QOE 4/2	-1.5	0.7	2.0	-0.4	20	100	30
-QOE 5/1	-1.5	0.7	2.0	-0.4	20	100	30
-QOE 5/1	-1.5	0.7	2.0	-0.4	20	100	30
-QOE 6/1	-1.5	0.7	2.0	-0.4	20	100	30
-QOE 6/2	-1.5	0.7	2.0	-0.4	20	100	30
LE 1/1	-1.5	0.7	2.0	-0.4	20	100	30
LE 1/2	-1.5	0.7	2.0	-0.4	20	100	30
-LE 2/1	-1.5	0.7	2.0	-0.4	20	100	30
-LE 2/2	-1.5	0.7	2.0	-0.4	20	100	30
-LE 3/1	-1.5	0.7	2.0	-0.4	20	100	30
-LE 3/2	-1.5	0.7	2.0	-0.4	20	100	30

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
D 0/1	-1.5	0.7	2.0	-0.42	40	120	45
D 1/1	-1.5	0.7	2.0	-0.42	40	120	45
D 2/1	-1.5	0.7	2.0	-0.42	40	120	45
D 3/1	-1.5	0.7	2.0	-0.42	40	120	45
D 4/1	-1.5	0.7	2.0	-0.42	40	120	45
D 5/1	-1.5	0.7	2.0	-0.42	40	120	45
D 6/1	-1.5	0.7	2.0	-0.42	40	120	45
D 7/1	-1.5	0.7	2.0	-0.42	40	120	45
D 8/1	-1.5	0.7	2.0	-0.42	40	120	45
D 9/1	-1.5	0.7	2.0	-0.42	40	120	45
D 10/1	-1.5	0.7	2.0	-0.42	40	120	45
D 11/1	-1.5	0.7	2.0	-0.42	40	120	45
D 12/1	-1.5	0.7	2.0	-0.42	40	120	45
D 13/1	-1.5	0.7	2.0	-0.42	40	120	45

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
D 14/1	-1.5	0.7	2.0	-0.42	40	120	45
D 15/1	-1.5	0.7	2.0	-0.42	40	120	45
D 0/2	-1.5	0.7	2.0	-0.42	40	120	45
D 1/2	-1.5	0.7	2.0	-0.42	40	120	45
D 2/2	-1.5	0.7	2.0	-0.42	40	120	45
D 3/2	-1.5	0.7	2.0	-0.42	40	120	45
D 4/2	-1.5	0.7	2.0	-0.42	40	120	45
D 5/2	-1.5	0.7	2.0	-0.42	40	120	45
D 6/2	-1.5	0.7	2.0	-0.42	40	120	45
D 7/2	-1.5	0.7	2.0	-0.42	40	120	45
D 8/2	-1.5	0.7	2.0	-0.42	40	120	45
D 9/2	-1.5	0.7	2.0	-0.42	40	120	45
D 10/2	-1.5	0.7	2.0	-0.42	40	120	45
D 11/2	-1.5	0.7	2.0	-0.42	40	120	45

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
D 12/2	-1.5	0.7	2.0	-0.42	40	120	45
D 13/2	-1.5	0.7	2.0	-0.42	40	120	45
D 14/2	-1.5	0.7	2.0	-0.42	40	120	45
D 15/2	-1.5	0.7	2.0	-0.42	40	120	45

6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh min (uA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Ios max (mA)	Cout max (pF)
Q 0/1	2.4	0.4	-3000	8.0	20	-20	-40.0	40
Q 1/1	2.4	0.4	-3000	8.0	20	-20	-40.0	40
Q 2/1	2.4	0.4	-3000	8.0	20	-20	-40.0	40
Q 3/1	2.4	0.4	-3000	8.0	20	-20	-40.0	40
Q 4/1	2.4	0.4	-3000	8.0	20	-20	-40.0	40
Q 5/1	2.4	0.4	-3000	8.0	20	-20	-40.0	40
Q 6/1	2.4	0.4	-3000	8.0	20	-20	-40.0	40

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh min (uA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Ios max (mA)	Cout max (pF)
Q 7/1	2.4	0.4	-3000	8.0	20	-20	-40.0	40
Q 8/1	2.4	0.4	-3000	8.0	20	-20	-40.0	40
Q 9/1	2.4	0.4	-3000	8.0	20	-20	-40.0	40
Q 10/1	2.4	0.4	-3000	8.0	20	-20	-40.0	40
Q 11/1	2.4	0.4	-3000	8.0	20	-20	-40.0	40
Q 12/1	2.4	0.4	-3000	8.0	20	-20	-40.0	40
Q 13/1	2.4	0.4	-3000	8.0	20	-20	-40.0	40
Q 14/1	2.4	0.4	-3000	8.0	20	-20	-40.0	40
Q 15/1	2.4	0.4	-3000	8.0	20	-20	-40.0	40
Q 0/2	2.4	0.4	-3000	8.0	20	-20	-40.0	40
Q 1/2	2.4	0.4	-3000	8.0	20	-20	-40.0	40
Q 2/2	2.4	0.4	-3000	8.0	20	-20	-40.0	40
Q 3/2	2.4	0.4	-3000	8.0	20	-20	-40.0	40
Q 4/2	2.4	0.4	-3000	8.0	20	-20	-40.0	40

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh min (uA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Ios max (mA)	Cout max (pF)
Q 5/2	2.4	0.4	-3000	8.0	20	-20	-40.0	40
Q 6/2	2.4	0.4	-3000	8.0	20	-20	-40.0	40
Q 7/2	2.4	0.4	-3000	8.0	20	-20	-40.0	40
Q 8/2	2.4	0.4	-3000	8.0	20	-20	-40.0	40
Q 9/2	2.4	0.4	-3000	8.0	20	-20	-40.0	40
Q 10/2	2.4	0.4	-3000	8.0	20	-20	-40.0	40
Q 11/2	2.4	0.4	-3000	8.0	20	-20	-40.0	40
Q 12/2	2.4	0.4	-3000	8.0	20	-20	-40.0	40
Q 13/2	2.4	0.4	-3000	8.0	20	-20	-40.0	40
Q 14/2	2.4	0.4	-3000	8.0	20	-20	-40.0	40
Q 15/2	2.4	0.4	-3000	8.0	20	-20	-40.0	40
D 0/1	2.4	0.4	-2980	8.0	40	-420	-40.1	45
D 1/1	2.4	0.4	-2980	8.0	40	-420	-40.1	45
D 2/1	2.4	0.4	-2980	8.0	40	-420	-40.1	45

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh min (uA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Ios max (mA)	Cout max (pF)
D 3/1	2.4	0.4	-2980	8.0	40	-420	-40.1	45
D 4/1	2.4	0.4	-2980	8.0	40	-420	-40.1	45
D 5/1	2.4	0.4	-2980	8.0	40	-420	-40.1	45
D 6/1	2.4	0.4	-2980	8.0	40	-420	-40.1	45
D 7/1	2.4	0.4	-2980	8.0	40	-420	-40.1	45
D 8/1	2.4	0.4	-2980	8.0	40	-420	-40.1	45
D 9/1	2.4	0.4	-2980	8.0	40	-420	-40.1	45
D 10/1	2.4	0.4	-2980	8.0	40	-420	-40.1	45
D 11/1	2.4	0.4	-2980	8.0	40	-420	-40.1	45
D 12/1	2.4	0.4	-2980	8.0	40	-420	-40.1	45
D 13/1	2.4	0.4	-2980	8.0	40	-420	-40.1	45
D 14/1	2.4	0.4	-2980	8.0	40	-420	-40.1	45
D 15/1	2.4	0.4	-2980	8.0	40	-420	-40.1	45
D 0/2	2.4	0.4	-2980	8.0	40	-420	-40.1	45

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh min (uA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Ios max (mA)	Cout max (pF)
D 1/2	2.4	0.4	-2980	8.0	40	-420	-40.1	45
D 2/2	2.4	0.4	-2980	8.0	40	-420	-40.1	45
D 3/2	2.4	0.4	-2980	8.0	40	-420	-40.1	45
D 4/2	2.4	0.4	-2980	8.0	40	-420	-40.1	45
D 5/2	2.4	0.4	-2980	8.0	40	-420	-40.1	45
D 6/2	2.4	0.4	-2980	8.0	40	-420	-40.1	45
D 7/2	2.4	0.4	-2980	8.0	40	-420	-40.1	45
D 8/2	2.4	0.4	-2980	8.0	40	-420	-40.1	45
D 9/2	2.4	0.4	-2980	8.0	40	-420	-40.1	45
D 10/2	2.4	0.4	-2980	8.0	40	-420	-40.1	45
D 11/2	2.4	0.4	-2980	8.0	40	-420	-40.1	45
D 12/2	2.4	0.4	-2980	8.0	40	-420	-40.1	45
D 13/2	2.4	0.4	-2980	8.0	40	-420	-40.1	45
D 14/2	2.4	0.4	-2980	8.0	40	-420	-40.1	45

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:06Z
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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh min (uA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Ios max (mA)	Cout max (pF)
D 15/2	2.4	0.4	-2980	8.0	40	-420	-40.1	45
-QOE	2.5	0.4	-360	3.6	NA	NA	-15.0	NA
PU	2.4	NA	-1910	NA	NA	NA	NA	NA

6.3.3 Dynamic parameters. The dynamic parameters appear with the dynamic test.

6.4 Module function. The DGJ module contains two independent 16-bit parallel registers. Each D-type register is divided into 8-bit groups with common load enable, read/write controls and individual clock and data output enables. The module is comprised of two bidirectional 16-bit write/read input buses (D Bus) and two 16-bit 3-state output buses (Q bus). A 1 KOhm pull-up resistor, whose output is pinned out, is available for logic "1" sourcing. The load logic enables the simultaneous loading on the rising edge of the clock input of both the lower (CLK 1/X) and upper (CLK 2/X) data bytes into the registers. All three load logic inputs (LE 1/X, -LE 2/X, -LE 3/X) must be active to perform a D Bus load (see figure 1). The Q Bus logic allows the lower or upper byte of data in the appropriate register to be placed onto the Q Bus. Selection of either lower or upper byte is controlled by which inputs to the Q Bus logic are active. For lower byte selection, inputs QOE 1/X, -QOE 2/X, and -QOE 3/X must all be active and for the upper byte, inputs QOE 4/X, -QOE 5/X, and -QOE 6/X must be active (see figure 1). The D Bus logic enables the data in the registers to be read by way of the D Bus. DOE 1/X, -DOE 2/X, and -DOE 3/X inputs must all be active to perform a D Bus read (see figure 1). The rising edge of the clock inputs, CLK 1/X and CLK 2/X triggers the loading of data into the lower and upper registers respectively.

6.4.1 Register read timing precaution. Caution must be taken to ensure that any system drivers connected to the D Bus (D 0/X to D 15/X) are never turned on while the DGJ's D Bus buffers are enabled (DOE 1/X, -DOE 2/X, and -DOE 3/X are all active). This prevents the possibility of having bus contention which generates a large amount of system noise.

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6.4.2 Crosstalk usage restrictions. To assure crosstalk levels below 400 mV, a maximum of 2 lines per nibble (4 bits; Q 0/X-Q 3/X; Q 4/X-Q 7/X) can be simultaneously switched. Up to all 8 nibbles can be switched simultaneously.

6.4.3 Signal state assignments. The information below describes the active state assignments for each control and data signal:

Electrical signal	Description	True electrical state
DOE 1/X	D Bus Output Enable Logic Input	High
-DOE 2/X	D Bus Output Enable Logic Input	Low
-DOE 3/X	D Bus Output Enable Logic Input	Low
QOE 1/X	Q Bus Lower Byte Output Enable Logic Input	High
-QOE 2/X	Q Bus Lower Byte Output Enable Logic Input	Low
-QOE 3/X	Q Bus Lower Byte Output Enable Logic Input	Low
QOE 4/X	Q Bus Upper Byte Output Enable Logic Input	High
-QOE 5/X	Q Bus Lower Byte Output Enable Logic Input	Low
-QOE 6/X	Q Bus Lower Byte Output Enable Logic Input	Low
LE 1/X	D Bus Load Logic Input	High
-LE 2/X	D Bus Load Logic Input	Low
-LE 3/X	D Bus Load Logic Input	Low
CLK 1/X	Lower Byte Register Clock	Positive Going Edge
D 0/X-D 15/X	D Bus Input/Output	X

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Electrical signal	Description	True electrical state
Q 0/X-Q 15/X	Q Bus Outputs	X
-QOE	Q Bus Outputs Enable	Low
PU	Pullup Output	High
CLK 2/X	Upper Byte Register Clock	Positive Going Edge

6.4.4 Description of I/O pins.

Signal name	Pin number	I/O	Functional description
DOE 1/1 DOE 1/2 -DOE 2/1 -DOE 2/2 -DOE 3/1 -DOE 3/2	52 80 20 49 70 99	I I I I I I	These D bus logic inputs, when all activated, allow the data in the lower and upper registers to be output to the D bus.
QOE 1/1 QOE 1/2 -QOE 2/1 -QOE 2/2 -QOE 3/1 -QOE 3/2	4 84 48 45 65 46	I I I I I I	These Q bus lower byte logic inputs, when all active, allow the data in the lower byte register to be placed onto the Q data bus.
QOE 4/1 QOE 4/2 -QOE 5/1 -QOE 5/2 -QOE 6/1 -QOE 6/2	53 83 19 97 69 47	I I I I I I	These Q bus upper byte logic inputs, when all active, allow the data in the upper byte register to be placed onto the Q data bus.
LE 1/1 LE 1/2 -LE 2/1 -LE 2/2 -LE 3/1 -LE 3/2	3 33 54 98 5 96	I I I I I I	These D bus register enables, when all active, enable both the lower and upper D bus registers, allowing data present on the D bus to be loaded into the registers on the rising edge of the clock inputs.

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6.4.4 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
CLK 1/1	2	I	These input signals load the lower data byte on the D bus into the lower byte register on its rising edge.
CLK 1/2	79	I	
CLK 2/1	72	I	These input signals load the upper data byte on the D bus into the upper byte register on its rising edge.
CLK 2/2	50	I	
Q 0/1	63	O	These signals make up the two 16-bit parallel unidirectional data buses (Q bus).
Q 1/1	13	O	
Q 2/1	62	O	
Q 3/1	12	O	
Q 4/1	7	O	
Q 5/1	56	O	
Q 6/1	6	O	
Q 7/1	55	O	
Q 8/1	17	O	
Q 9/1	18	O	
Q 10/1	68	O	
Q 11/1	67	O	
Q 12/1	64	O	
Q 13/1	14	O	
Q 14/1	16	O	
Q 15/1	66	O	
Q 0/2	87	O	
Q 1/2	37	O	
Q 2/2	86	O	
Q 3/2	35	O	
Q 4/2	82	O	
Q 5/2	32	O	
Q 6/2	81	O	
Q 7/2	34	O	
Q 8/2	39	O	
Q 9/2	89	O	
Q 10/2	93	O	
Q 11/2	90	O	
Q 12/2	38	O	
Q 13/2	88	O	

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6.4.4 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
Q 14/2	36	O	These signals make up the two 16-bit parallel unidirectional data buses (Q bus).
Q 15/2	85	O	
-QOE	100	O	This is the Q bus upper byte output enable for Q 8/2 - Q 15/2.
D 0/1	57	I/O	These signals make up the two 16-bit parallel bidirectional data buses (D bus).
D 1/1	8	I/O	
D 2/1	58	I/O	
D 3/1	9	I/O	
D 4/1	59	I/O	
D 5/1	60	I/O	
D 6/1	11	I/O	
D 7/1	61	I/O	
D 8/1	73	I/O	
D 9/1	22	I/O	
D 10/1	71	I/O	
D 11/1	21	I/O	
D 12/1	23	I/O	
D 13/1	24	I/O	
D 14/1	74	I/O	
D 15/1	75	I/O	
D 0/2	30	I/O	
D 1/2	78	I/O	
D 2/2	28	I/O	
D 3/2	77	I/O	
D 4/2	27	I/O	
D 5/2	26	I/O	
D 6/2	76	I/O	
D 7/2	29	I/O	
D 8/2	91	I/O	
D 9/2	41	I/O	
D 10/2	92	I/O	
D 11/2	42	I/O	
D 12/2	95	I/O	
D 13/2	43	I/O	
D 14/2	94	I/O	
D 15/2	44	I/O	

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6.4.4. Description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
PU	51	O	1 KOhm pull-up resistor. This is a logic "1" source for pulling up unused module pins.

6.5 Description of tests. The following tables and associated figures describe the tests to be performed.

(a) Isolation test - table II.

(b) Continuity test - table III.

(c) Vcc stress test - table IV.

(d) Ibd test - table V.

(e) Vic test - table VI.

(f) Ios test - table VII.

(g) Iih test - table VIII.

(h) Iil test - table IX.

(i) Voh test - table X, figure 8.

(j) Vol test - table XI, figure 8.

(k) Iozh test - table XII.

(l) Iozl test - table XIII.

(m) Icc test - table XIV.

(n) Functional test - table XV, figure 9.

(o) Dynamic test - tables XVI and XVII, figures 10 and 11.

(p) PSFC test - table XVIII.

(q) PSTA test - table XIX, figure 12.

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(r) Output crosstalk tests - tables XX and XXI, figures 13 and 14.

(s) Output crosstalk duration test - table XXII, figure 15.

(t) Cin test - table XXIII.

(u) Cout test - table XXIV.

(v) Threshold test - table XXV.

(w) Disable time test - table XXVI, figures 16 through 19.

(x) Noise margin test - table XXVII, figures 20 and 21.

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6.6 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT, AV
Navy - AS, MC
Air Force - 13, 17, 19
DLA - ES

Preparing activity:
Navy - SH

(Project 5963-0060)

Agent: NW

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TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A	Isolation, frame	All pins	Frame	10	INF	MOhm	1-7

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A
Isolation measurement (see note 6)	NA NA	± 500 KOhm ± 50 KOhm	A A

2. The megohm bridge used shall be item R06.

3. In the setup conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, and 'B' shall correspond to the negative terminal.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. The minimum end-of-life limit is 1 MOhm.

7. Abbreviations used are defined in 6.1.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, Vcc	1	31	0.0	1.0	Ohm	1-4
B2	Continuity, GND	10	15				
B3	Continuity, GND	15	25				
B4	Continuity, GND	25	40	0.0	1.0	Ohm	1-4
B5	Continuity, Vcc	1	31	0.0	0.1	Ohm	1-4
B6	Continuity, GND	10	15				
B7	Continuity, GND	15	25				
B8	Continuity, GND	25	40	0.0	0.1	Ohm	1-4

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA NA	± 50 mOhm ± 5 mOhm	B1-B4 B5-B8

2. The ohmmeter used shall be item R10.

3. In the setup conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. Abbreviations used are defined in 6.1.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31	10,15 25,40	Module shall pass all remaining tests.	1-5

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	878 mA	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. Abbreviations used are defined in 6.1.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	DOE 1/1	52	5.5	0	100	uA	1-3
D2	-DOE 2/1	20					
D3	-DOE 3/1	70					
D4	DOE 1/2	80					
D5	-DOE 2/2	49					
D6	-DOE 3/2	99					
D7	-QOE 2/1	48					
D8	-QOE 3/1	65					
D9	QOE 1/1	4					
D10	QOE 4/1	53					
D11	-QOE 5/1	19					
D12	-QOE 6/1	69					
D13	-QOE 3/2	46					
D14	-QOE 2/2	45					
D15	QOE 1/2	84					
D16	QOE 4/2	83					
D17	-QOE 5/2	97					
D18	-QOE 6/2	47					
D19	LE 1/1	3					
D20	-LE 2/1	54	5.5	0	100	uA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D21	-LE 3/1	5	5.5	0	100	uA	1-3
D22	LE 1/2	33					
D23	-LE 2/2	98					
D24	-LE 3/2	96					
D25	CLK 1/1	2					
D26	CLK 2/1	72					
D27	CLK 1/2	79					
D28	CLK 2/2	50	5.5	0	100	uA	1-3
D29	D 0/1	57	5.5	0	120	uA	1-3
D30	D 1/1	8					
D31	D 2/1	58					
D32	D 3/1	9					
D33	D 4/1	59					
D34	D 5/1	60					
D35	D 6/1	11					
D36	D 7/1	61					
D37	D 8/1	73					
D38	D 9/1	22					
D39	D 10/1	71					
D40	D 11/1	21	5.5	0	120	uA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D41	D 12/1	23	5.5	0	120	uA	1-3
D42	D 13/1	24					
D43	D 14/1	74					
D44	D 15/1	75					
D45	D 0/2	30					
D46	D 1/2	78					
D47	D 2/2	28					
D48	D 3/2	77					
D49	D 4/2	27					
D50	D 5/2	26					
D51	D 6/2	76					
D52	D 7/2	29					
D53	D 8/2	91					
D54	D 9/2	41					
D55	D 10/2	92					
D56	D 11/2	42					
D57	D 12/2	95					
D58	D 13/2	43					
D59	D 14/2	94					
D60	D 15/2	44	5.5	0	120	uA	1-3

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NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D60
Power supply current limit	0.878 A	NA	D1-D60
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D60
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 D1-D60 D1-D60
Input current measurement	 NA NA	 ± 5 uA ± 6 uA	 D1-D28 D29-D60

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	DOE 1/1	52	-18.0	-1.5	-0.4	V	1-3
E2	-DOE 2/1	20					
E3	-DOE 3/1	70					
E4	DOE 1/2	80					
E5	-DOE 2/2	49					
E6	-DOE 3/2	99					
E7	-QOE 2/1	48					
E8	-QOE 3/1	65					
E9	QOE 1/1	4					
E10	QOE 4/1	53					
E11	-QOE 5/1	19					
E12	-QOE 6/1	69					
E13	-QOE 3/2	46					
E14	-QOE 2/2	45					
E15	QOE 1/2	84					
E16	QOE 4/2	83					
E17	-QOE 5/2	97					
E18	-QOE 6/2	47					
E19	LE 1/1	3					
E20	-LE 2/1	54	-18.0	-1.5	-0.4	V	1-3

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TABLE VI. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
E21	-LE 3/1	5	-18.0	-1.5	-0.4	V	1-3
E22	LE 1/2	33					
E23	-LE 2/2	98					
E24	-LE 3/2	96					
E25	CLK 1/1	2					
E26	CLK 2/1	72					
E27	CLK 1/2	79					
E28	CLK 2/2	50					
E29	D 0/1	57					
E30	D 1/1	8					
E31	D 2/1	58					
E32	D 3/1	9					
E33	D 4/1	59					
E34	D 5/1	60					
E35	D 6/1	11					
E36	D 7/1	61					
E37	D 8/1	73					
E38	D 9/1	22					
E39	D 10/1	71					
E40	D 11/1	21	-18.0	-1.5	-0.4	V	1-3

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TABLE VI. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Initial limits Max	Units	Notes
E41	D 12/1	23	-18.0	-1.5	-0.4	V	1-3
E42	D 13/1	24					
E43	D 14/1	74					
E44	D 15/1	75					
E45	D 0/2	30					
E46	D 1/2	78					
E47	D 2/2	28					
E48	D 3/2	77					
E49	D 4/2	27					
E50	D 5/2	26					
E51	D 6/2	76					
E52	D 7/2	29					
E53	D 8/2	91					
E54	D 9/2	41					
E55	D 10/2	92					
E56	D 11/2	42					
E57	D 12/2	95					
E58	D 13/2	43					
E59	D 14/2	94					
E60	D 15/2	44	-18.0	-1.5	-0.4	V	1-3

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NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	E1-E60
Power supply current limit	0.878 A	NA	E1-E60
Forcing current, (Iin)	-18 mA	± 1.8 mA	E1-E60
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 E1-E60 E1-E60
Input voltage measurement	NA	± 110 mV	E1-E60

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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NOTES: - Continued.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table VI and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE VII. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	Q 0/1	63	0.0	-225.0	-40.0	mA	1-4
F2	Q 1/1	13					
F3	Q 2/1	62					
F4	Q 3/1	12					
F5	Q 4/1	7					
F6	Q 5/1	56					
F7	Q 6/1	6					
F8	Q 7/1	55					
F9	Q 8/1	17					
F10	Q 9/1	18					
F11	Q 10/1	68					
F12	Q 11/1	67					
F13	Q 12/1	64					
F14	Q 13/1	14					
F15	Q 14/1	16					
F16	Q 15/1	66					
F17	Q 0/2	87					
F18	Q 1/2	37					
F19	Q 2/2	86					
F20	Q 3/2	35	0.0	-225.0	-40.0	mA	1-4

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TABLE VII. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F21	Q 4/2	82	0.0	-225.0	-40.0	mA	1-4
F22	Q 5/2	32					
F23	Q 6/2	81					
F24	Q 7/2	34					
F25	Q 8/2	39					
F26	Q 9/2	89					
F27	Q 10/2	93					
F28	Q 11/2	90					
F29	Q 12/2	38					
F30	Q 13/2	88					
F31	Q 14/2	36					
F32	Q 15/2	85	0.0	-225.0	-40.0	mA	1-4
F33	D 0/1	57	0.0	-225.1	-40.1	mA	1-4
F34	D 1/1	8					
F35	D 2/1	58					
F36	D 3/1	9					
F37	D 4/1	59					
F38	D 5/1	60					
F39	D 6/1	11					
F40	D 7/1	61	0.0	-225.1	-40.1	mA	1-4

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TABLE VII. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F41	D 8/1	73	0.0	-225.1	-40.1	mA	1-4
F42	D 9/1	22					
F43	D 10/1	71					
F44	D 11/1	21					
F45	D 12/1	23					
F46	D 13/1	24					
F47	D 14/1	74					
F48	D 15/1	75					
F49	D 0/2	30					
F50	D 1/2	78					
F51	D 2/2	28					
F52	D 3/2	77					
F53	D 4/2	27					
F54	D 5/2	26					
F55	D 6/2	76					
F56	D 7/2	29					
F57	D 8/2	91					
F58	D 9/2	41					
F59	D 10/2	92					
F60	D 11/2	42	0.0	-225.1	-40.1	mA	1-4

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TABLE VII. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F61	D 12/2	95	0.0	-225.1	-40.1	mA	1-4
F62	D 13/2	43					
F63	D 14/2	94					
F64	D 15/2	44	0.0	-225.1	-40.1	mA	1-4
F65	-QOE	100	0.0	-100.4	-15.0	mA	1-4

NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F65
Power supply current limit	0.878 A	NA	F1-F65
Forcing voltage, (Vout)	0.0 V	± 20 mV	F1-F65
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	F1-F65 F1-F65
Output current measurement	NA NA NA	± 9.25 mA ± 9.26 mA ± 4.27 mA	F1-F32 F33-F64 F65

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
G1	DOE 1/1	52	2.7	0	20	uA	1-3
G2	-DOE 2/1	20					
G3	-DOE 3/1	70					
G4	DOE 1/2	80					
G5	-DOE 2/2	49					
G6	-DOE 3/2	99					
G7	-QOE 2/1	48					
G8	-QOE 3/1	65					
G9	QOE 1/1	4					
G10	QOE 4/1	53					
G11	-QOE 5/1	19					
G12	-QOE 6/1	69					
G13	-QOE 3/2	46					
G14	-QOE 2/2	45					
G15	QOE 1/2	84					
G16	QOE 4/2	83					
G17	-QOE 5/2	97					
G18	-QOE 6/2	47					
G19	LE 1/1	3					
G20	-LE 2/1	54	2.7	0	20	uA	1-3

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TABLE VIII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
G21	-LE 3/1	5	2.7	0	20	uA	1-3
G22	LE 1/2	33					
G23	-LE 2/2	98					
G24	-LE 3/2	96					
G25	CLK 1/1	2					
G26	CLK 2/1	72					
G27	CLK 1/2	79					
G28	CLK 2/2	50	2.7	0	20	uA	1-3
G29	D 0/1	57	2.7	-20	40	uA	1-3
G30	D 1/1	8					
G31	D 2/1	58					
G32	D 3/1	9					
G33	D 4/1	59					
G34	D 5/1	60					
G35	D 6/1	11					
G36	D 7/1	61					
G37	D 8/1	73					
G38	D 9/1	22					
G39	D 10/1	71					
G40	D 11/1	21	2.7	-20	40	uA	1-3

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TABLE VIII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G41	D 12/1	23	2.7	-20	40	uA	1-3
G42	D 13/1	24					
G43	D 14/1	74					
G44	D 15/1	75					
G45	D 0/2	30					
G46	D 1/2	78					
G47	D 2/2	28					
G48	D 3/2	77					
G49	D 4/2	27					
G50	D 5/2	26					
G51	D 6/2	76					
G52	D 7/2	29					
G53	D 8/2	91					
G54	D 9/2	41					
G55	D 10/2	92					
G56	D 11/2	42					
G57	D 12/2	95					
G58	D 13/2	43					
G59	D 14/2	94					
G60	D 15/2	44	2.7	-20	40	uA	1-3

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NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G60
Power supply current limit	0.878 A	NA	G1-G60
Forcing voltage, (Vih)	2.7 V	± 135 mV	G1-G60
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	G1-G60 G1-G60
Input current measurement	NA NA	± 1 uA ± 3 uA	G1-G28 G29-G60

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Initial limits Max	Units	Notes
H1	DOE 1/1	52	0.4	-400	-30	uA	1-3
H2	-DOE 2/1	20					
H3	-DOE 3/1	70					
H4	DOE 1/2	80					
H5	-DOE 2/2	49					
H6	-DOE 3/2	99					
H7	-QOE 2/1	48					
H8	-QOE 3/1	65					
H9	QOE 1/1	4					
H10	QOE 4/1	53					
H11	-QOE 5/1	19					
H12	-QOE 6/1	69					
H13	-QOE 3/2	46					
H14	-QOE 2/2	45					
H15	QOE 1/2	84					
H16	QOE 4/2	83					
H17	-QOE 5/2	97					
H18	-QOE 6/2	47					
H19	LE 1/1	3					
H20	-LE 2/1	54	0.4	-400	-30	uA	1-3

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TABLE IX. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H21	-LE 3/1	5	0.4	-400	-30	uA	1-3
H22	LE 1/2	33					
H23	-LE 2/2	98					
H24	-LE 3/2	96					
H25	CLK 1/1	2					
H26	CLK 2/1	72					
H27	CLK 1/2	79					
H28	CLK 2/2	50	0.4	-400	-30	uA	1-3
H29	D 0/1	57	0.4	-420	-80	uA	1-3
H30	D 1/1	8					
H31	D 2/1	58					
H32	D 3/1	9					
H33	D 4/1	59					
H34	D 5/1	60					
H35	D 6/1	11					
H36	D 7/1	61					
H37	D 8/1	73					
H38	D 9/1	22					
H39	D 10/1	71					
H40	D 11/1	21	0.4	-420	-80	uA	1-3

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Check the source to verify that this is the current version before use.

TABLE IX. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Initial limits Max	Units	Notes
H41	D 12/1	23	0.4	-420	-80	uA	1-3
H42	D 13/1	24					
H43	D 14/1	74					
H44	D 15/1	75					
H45	D 0/2	30					
H46	D 1/2	78					
H47	D 2/2	28					
H48	D 3/2	77					
H49	D 4/2	27					
H50	D 5/2	26					
H51	D 6/2	76					
H52	D 7/2	29					
H53	D 8/2	91					
H54	D 9/2	41					
H55	D 10/2	92					
H56	D 11/2	42					
H57	D 12/2	95					
H58	D 13/2	43					
H59	D 14/2	94					
H60	D 15/2	44	0.4	-420	-80	uA	1-3

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NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	H1-H60
Power supply current limit	0.878 A	NA	H1-H60
Forcing voltage, (Vil)	0.4 V	± 20 mV	H1-H60
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	H1-H60 H1-H60
Input current measurement	NA NA	± 18.5 uA ± 17.0 uA	H1-H28 H29-H60

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table IX and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE X. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J1	Q 0/1	63	-3.0	2.4	4.5	V	1-4
J2	Q 1/1	13					
J3	Q 2/1	62					
J4	Q 3/1	12					
J5	Q 4/1	7					
J6	Q 5/1	56					
J7	Q 6/1	6					
J8	Q 7/1	55					
J9	Q 8/1	17					
J10	Q 9/1	18					
J11	Q 10/1	68					
J12	Q 11/1	67					
J13	Q 12/1	64					
J14	Q 13/1	14					
J15	Q 14/1	16					
J16	Q 15/1	66					
J17	Q 0/2	87					
J18	Q 1/2	37					
J19	Q 2/2	86					
J20	Q 3/2	35	-3.0	2.4	4.5	V	1-4

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TABLE X. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J21	Q 4/2	82	-3.0	2.4	4.5	V	1-4
J22	Q 5/2	32					
J23	Q 6/2	81					
J24	Q 7/2	34					
J25	Q 8/2	39					
J26	Q 9/2	89					
J27	Q 10/2	93					
J28	Q 11/2	90					
J29	Q 12/2	38					
J30	Q 13/2	88					
J31	Q 14/2	36					
J32	Q 15/2	85	-3.0	2.4	4.5	V	1-4
J33	D 0/1	57	-2.98	2.4	4.5	V	1-4
J34	D 1/1	8					
J35	D 2/1	58					
J36	D 3/1	9					
J37	D 4/1	59					
J38	D 5/1	60					
J39	D 6/1	11					
J40	D 7/1	61	-2.98	2.4	4.5	V	1-4

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TABLE X. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J41	D 8/1	73	-2.98	2.4	4.5	V	1-4
J42	D 9/1	22					
J43	D 10/1	71					
J44	D 11/1	21					
J45	D 12/1	23					
J46	D 13/1	24					
J47	D 14/1	74					
J48	D 15/1	75					
J49	D 0/2	30					
J50	D 1/2	78					
J51	D 2/2	28					
J52	D 3/2	77					
J53	D 4/2	27					
J54	D 5/2	26					
J55	D 6/2	76					
J56	D 7/2	29					
J57	D 8/2	91					
J58	D 9/2	41					
J59	D 10/2	92					
J60	D 11/2	42	-2.98	2.4	4.5	V	1-4

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TABLE X. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J61	D 12/2	95	-2.98	2.4	4.5	V	1-4
J62	D 13/2	43					
J63	D 14/2	94					
J64	D 15/2	44	-2.98	2.4	4.5	V	1-4
J65	-QOE	100	-0.36	2.5	4.5	V	1-4
J66	PU	51	-1.91	2.4	2.78	V	1-4

NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J66
Power supply current limit	0.878 A	NA	J1-J66
Forcing current, (Ioh)	-3.0 mA -2.98 mA -0.36 mA -1.91 mA	± 30.0 uA ± 29.8 uA ± 3.6 uA ± 19.1 uA	J1-J32 J33-J64 J65 J66
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	J1-J66 J1-J66
Output voltage measurement	NA NA NA	± 105 mV ± 100 mV ± 19 mV	J1-J64 J65 J66

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NOTES: - Continued

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

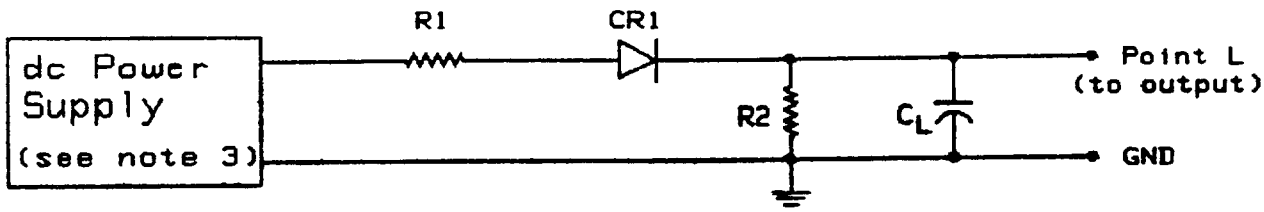
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 8.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) CL = 150 ± 15 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) CL = 50 ± 5 pF (including probe and parasitic capacitance) for crosstalk and disable time tests.

(c) CL = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(d) The functional test shall be performed with only a 150 ± 15 pF load (including probe and parasitic capacitance) attached to each output.

(e) CL not required for any other tests.

3. The load power supply voltage shall be 2.1 V ± 1 percent for the output disable time test and 2.5 V ± 1 percent for all other tests.

FIGURE 8. Logic load circuit.

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NOTES: - Continued.

4. Select R1 for a current flow of Iol when point L is held at Vol with the load power supply set at 2.5 V ± 1 percent. The values of Iol and Vol for each output are listed below.

5. The value of R2 for each output load shall be as specified below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
6,7,12,13,14,16, 17,18,32,34,35,36, 37,38,39,55,56,62, 63,64,66,67,68,81, 82,85,86,87,88,89, 90,93	800	8.0	0.4
8,9,11,21,22,23, 24,26,27,28,29,30, 41,42,43,44,57,58, 59,60,61,71,73,74, 75,76,77,78,91,92, 94,95	805	8.0	0.4
51	1257	NA	NA
100	6944	3.6	0.4

1/ All resistors are metal film, ± 1 percent, 1/4 W.

6. The load power supply voltage shall be 2.1Vdc ± 1 percent for the disable time test and 2.5Vdc ± 1 percent for all other tests.

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TABLE XI. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
K1	Q 0/1	63	8.0	0.0	0.4	V	1-4
K2	Q 1/1	13					
K3	Q 2/1	62					
K4	Q 3/1	12					
K5	Q 4/1	7					
K6	Q 5/1	56					
K7	Q 6/1	6					
K8	Q 7/1	55					
K9	Q 8/1	17					
K10	Q 9/1	18					
K11	Q 10/1	68					
K12	Q 11/1	67					
K13	Q 12/1	64					
K14	Q 13/1	14					
K15	Q 14/1	16					
K16	Q 15/1	66					
K17	Q 0/2	87					
K18	Q 1/2	37					
K19	Q 2/2	86					
K20	Q 3/2	35	8.0	0.0	0.4	V	1-4

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TABLE XI. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
K21	Q 4/2	82	8.0	0.0	0.4	V	1-4
K22	Q 5/2	32					
K23	Q 6/2	81					
K24	Q 7/2	34					
K25	Q 8/2	39					
K26	Q 9/2	89					
K27	Q 10/2	93					
K28	Q 11/2	90					
K29	Q 12/2	38					
K30	Q 13/2	88					
K31	Q 14/2	36					
K32	Q 15/2	85					
K33	D 0/1	57					
K34	D 1/1	8					
K35	D 2/1	58					
K36	D 3/1	9					
K37	D 4/1	59					
K38	D 5/1	60					
K39	D 6/1	11					
K40	D 7/1	61	8.0	0.0	0.4	V	1-4

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TABLE XI. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
K41	D 8/1	73	8.0	0.0	0.4	V	1-4
K42	D 9/1	22					
K43	D 10/1	71					
K44	D 11/1	21					
K45	D 12/1	23					
K46	D 13/1	24					
K47	D 14/1	74					
K48	D 15/1	75					
K49	D 0/2	30					
K50	D 1/2	78					
K51	D 2/2	28					
K52	D 3/2	77					
K53	D 4/2	27					
K54	D 5/2	26					
K55	D 6/2	76					
K56	D 7/2	29					
K57	D 8/2	91					
K58	D 9/2	41					
K59	D 10/2	92					
K60	D 11/2	42	8.0	0.0	0.4	V	1-4

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TABLE XI. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
K61	D 12/2	95	8.0	0.0	0.4	V	1-4
K62	D 13/2	43					
K63	D 14/2	94					
K64	D 15/2	44	8.0				
K65	-QOE	100	3.6	0.0	0.4	V	1-4

NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	K1-K65
Power supply current limit	0.878 A	NA	K1-K65
Forcing current, (Iol)	8.0 mA 3.6 mA	± 80 uA ± 36 uA	K1-K64 K65
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	K1-K65 K1-K65
Output voltage measurement	NA	± 20 mV	K1-K65

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XI and the voltage then measured.

3. The load shall be as indicated on figure 8.

4. Abbreviations used are defined in 6.1.

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TABLE XII. Output leakage current, high level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	Q 0/1	63	2.7	-20	20	uA	1-3
L2	Q 1/1	13					
L3	Q 2/1	62					
L4	Q 3/1	12					
L5	Q 4/1	7					
L6	Q 5/1	56					
L7	Q 6/1	6					
L8	Q 7/1	55					
L9	Q 8/1	17					
L10	Q 9/1	18					
L11	Q 10/1	68					
L12	Q 11/1	67					
L13	Q 12/1	64					
L14	Q 13/1	14					
L15	Q 14/1	16					
L16	Q 15/1	66					
L17	Q 0/2	87					
L18	Q 1/2	37					
L19	Q 2/2	86					
L20	Q 3/2	35	2.7	-20	20	uA	1-3

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TABLE XII. Output leakage current, high level, test - Continued.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
L21	Q 4/2	82	2.7	-20	20	uA	1-3
L22	Q 5/2	32					
L23	Q 6/2	81					
L24	Q 7/2	34					
L25	Q 8/2	39					
L26	Q 9/2	89					
L27	Q 10/2	93					
L28	Q 11/2	90					
L29	Q 12/2	38					
L30	Q 13/2	88					
L31	Q 14/2	36					
L32	Q 15/2	85	2.7	-20	20	uA	1-3
L33	D 0/1	57	2.7	-20	40	uA	1-3
L34	D 1/1	8					
L35	D 2/1	58					
L36	D 3/1	9					
L37	D 4/1	59					
L38	D 5/1	60					
L39	D 6/1	11					
L40	D 7/1	61	2.7	-20	40	uA	1-3

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TABLE XII. Output leakage current, high level, test - Continued.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
L41	D 8/1	73	2.7	-20	40	uA	1-3
L42	D 9/1	22					
L43	D 10/1	71					
L44	D 11/1	21					
L45	D 12/1	23					
L46	D 13/1	24					
L47	D 14/1	74					
L48	D 15/1	75					
L49	D 0/2	30					
L50	D 1/2	78					
L51	D 2/2	28					
L52	D 3/2	77					
L53	D 4/2	27					
L54	D 5/2	26					
L55	D 6/2	76					
L56	D 7/2	29					
L57	D 8/2	91					
L58	D 9/2	41					
L59	D 10/2	92					
L60	D 11/2	42	2.7	-20	40	uA	1-3

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TABLE XII. Output leakage current, high level, test - Continued.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
L61	D 12/2	95	2.7	-20	40	uA	1-3
L62	D 13/2	43					
L63	D 14/2	94					
L64	D 15/2	44	2.7	-20	40	uA	1-3

NOTES:

1. Output leakage current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	L1-L64
Power supply current limit	0.878 A	NA	L1-L64
Forcing voltage, (Voh)	2.7 V	± 135 mV	L1-L64
Pattern voltage high, (VH)	3.0 V	± 100 mV	L1-L64
low, (VL)	0.0 V	± 100 mV	L1-L64
Output current measurement	NA	± 2 uA	L1-L32
	NA	± 3 uA	L33-L64

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Output leakage current, low level, test.

Test cond	Signal name	Pin no.	Vol (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
M1	Q 0/1	63	0.4	-20	20	uA	1-3
M2	Q 1/1	13					
M3	Q 2/1	62					
M4	Q 3/1	12					
M5	Q 4/1	7					
M6	Q 5/1	56					
M7	Q 6/1	6					
M8	Q 7/1	55					
M9	Q 8/1	17					
M10	Q 9/1	18					
M11	Q 10/1	68					
M12	Q 11/1	67					
M13	Q 12/1	64					
M14	Q 13/1	14					
M15	Q 14/1	16					
M16	Q 15/1	66					
M17	Q 0/2	87					
M18	Q 1/2	37					
M19	Q 2/2	86					
M20	Q 3/2	35	0.4	-20	20	uA	1-3

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TABLE XIII. Output leakage current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vol (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
M21	Q 4/2	82	0.4	-20	20	uA	1-3
M22	Q 5/2	32					
M23	Q 6/2	81					
M24	Q 7/2	34					
M25	Q 8/2	39					
M26	Q 9/2	89					
M27	Q 10/2	93					
M28	Q 11/2	90					
M29	Q 12/2	38					
M30	Q 13/2	88					
M31	Q 14/2	36					
M32	Q 15/2	85	0.4	-20	20	uA	1-3
M33	D 0/1	57	0.4	-420	-80	uA	1-3
M34	D 1/1	8					
M35	D 2/1	58					
M36	D 3/1	9					
M37	D 4/1	59					
M38	D 5/1	60					
M39	D 6/1	11					
M40	D 7/1	61	0.4	-420	-80	uA	1-3

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Check the source to verify that this is the current version before use.

TABLE XIII. Output leakage current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vol (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
M41	D 8/1	73	0.4	-420	-80	uA	1-3
M42	D 9/1	22					
M43	D 10/1	71					
M44	D 11/1	21					
M45	D 12/1	23					
M46	D 13/1	24					
M47	D 14/1	74					
M48	D 15/1	75					
M49	D 0/2	30					
M50	D 1/2	78					
M51	D 2/2	28					
M52	D 3/2	77					
M53	D 4/2	27					
M54	D 5/2	26					
M55	D 6/2	76					
M56	D 7/2	29					
M57	D 8/2	91					
M58	D 9/2	41					
M59	D 10/2	92					
M60	D 11/2	42	0.4	-420	-80	uA	1-3

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TABLE XIII. Output leakage current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vol (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
M61	D 12/2	95	0.4	-420	-80	uA	1-3
M62	D 13/2	43					
M63	D 14/2	94					
M64	D 15/2	44	0.4	-420	-80	uA	1-3

NOTES:

1. Output leakage current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	M1-M64
Power supply current limit	0.878 A	NA	M1-M64
Forcing voltage, (Vol)	0.4 V	± 20 mV	M1-M64
Pattern voltage high, (VH)	3.0 V	± 100 mV	M1-M64
low, (VL)	0.0 V	± 100 mV	M1-M64
Output current measurement	NA	± 2 uA	M1-M32
	NA	± 17 uA	M33-M64

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Icc	1,31	30	585	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	0.878 A	NA	N
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 N N
Power supply current measurement	NA	± 27.75 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate	1-3
P2	Functional test maximum power supply voltage maximum pattern rate	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	0.878 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 120 mV ± 20 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 20 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 8). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 9.

3. Abbreviations used are defined in 6.1.

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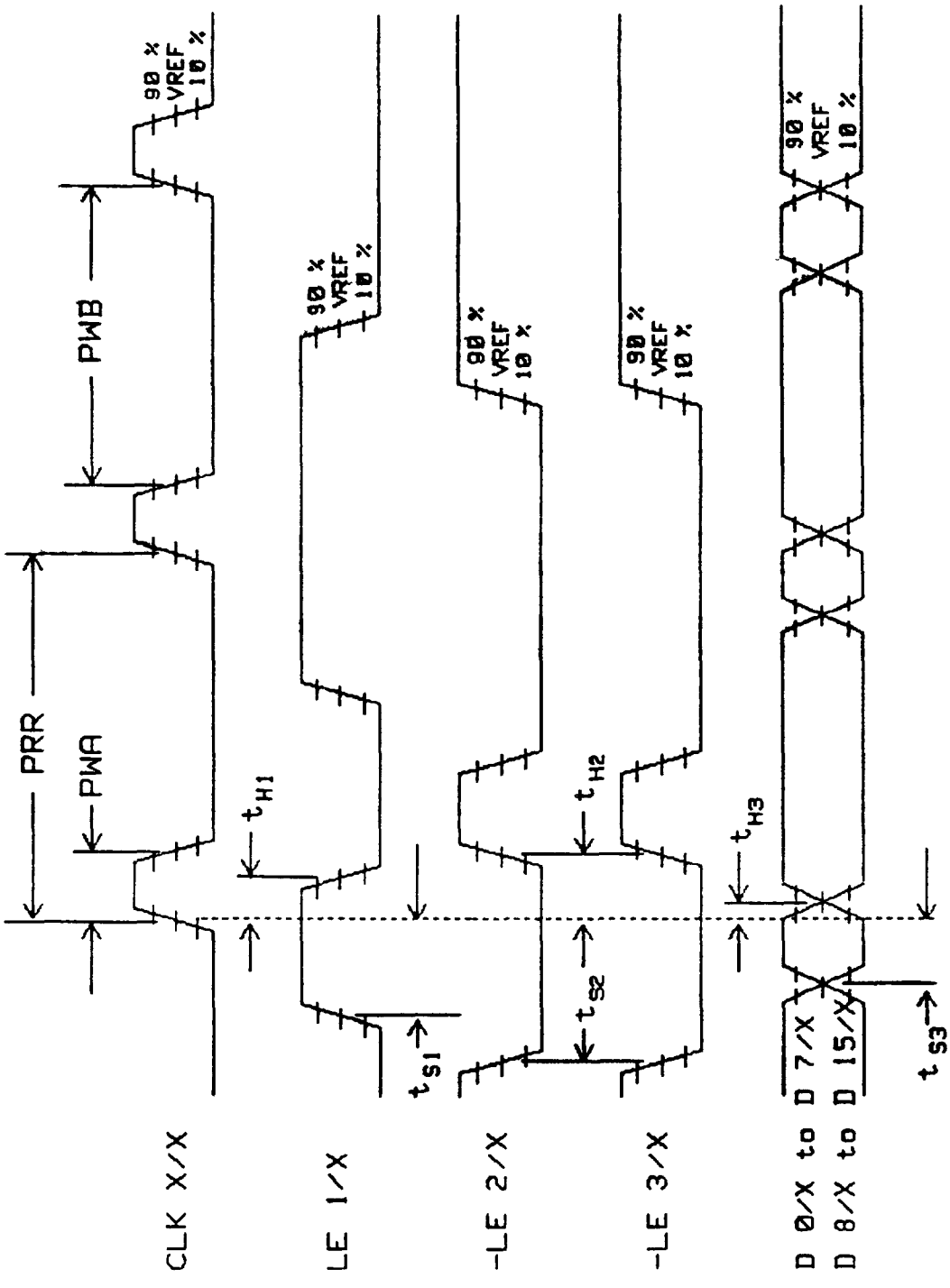


FIGURE 9. Input timing parameters for table XV.

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NOTES:

1. Input parameters for figure 9:

PRR = 83.3 nS minimum.
PWA = 20 nS minimum.
PWB = 20 nS minimum.
TH1 = 3 nS minimum.
TH2 = 1 nS minimum.
TH3 = 5 nS minimum.
TS1 = 49 nS minimum.
TS2 = 79 uS minimum.
TS3 = 20 uS minimum.
VREF = 1.3 $\pm$ 0.1 V.

2. Pattern voltages shall be as specified in the test tolerance table.

3. Rise times (as measured between the 10 percent and 90 percent points) shall be 15 nS maximum.

4. Fall times (as measured between the 10 percent and 90 percent points) shall be 6 nS maximum.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate minimum power supply	1-7
Q2	Dynamic test, sample maximum pattern rate maximum power supply	1-7
Q3	Dynamic test, qualification maximum pattern rate minimum power supply	1-7
Q4	Dynamic test, qualification maximum pattern rate maximum power supply	1-7

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	0.878 A	NA	Q1-Q4
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 120 mV ± 20 mV	 Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 105 mV	Q1,Q3
high, (Voh)	NA	± 155 mV	Q2,Q4
low, (Vol)	NA	± 20 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) The output compare levels shall be as follows:

High level: $2.4 \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.4$

3. The load circuit for this test is specified on figure 8.

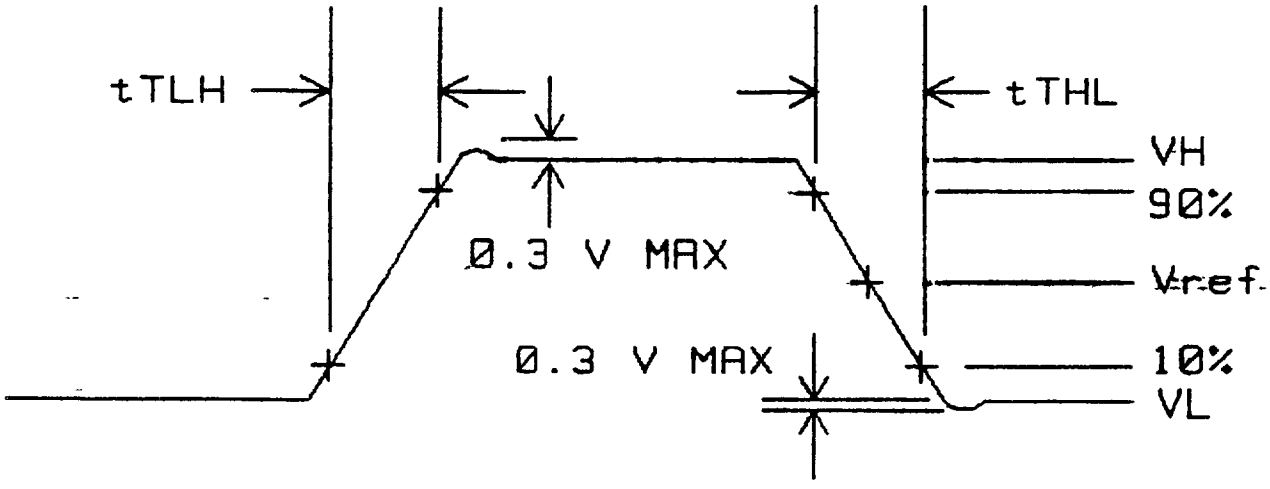
4. All of the input waveforms shall meet the requirements shown on figure 10.

5. This test shall be performed as indicated in table XVI. The dynamic parameters are found in table XVII.

6. Input/Output (I/O) waveforms are as referenced on figure 11.

7. Abbreviations used are defined in 6.1.

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NOTES:

1. VH and VL are as specified in the dynamic test tolerance table.

2. Vref = 1.5 V.

3. Rise and fall times for the above waveform are as follows:

Input pin	tTLH	tTHL
52,20,70,48,65,4,3,54,5,19,69, 63,80,49,99,46,45,84,33,98,96, 97,47,83,57,8,58,9,59,60,11, 61,2,73,22,71,21,23,24,74,75, 72,30,78,28,77,27,26,76,29,79, 91,41,92,42,95,43,94,44,50	15.0 ± 1.5 nS	6.0 ± 1.5 nS

FIGURE 10. Sample input waveform for table XVI.

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TABLE XVII. Dynamic parameters.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
12	1-16	CLK 1/1, CLK 2/1 to Q 0/1-Q 7/1, Q 8/1-Q 15/1 tPHL	-	-	-	73	58	73	nS	11a
12	17-32	CLK 1/1, CLK 2/1 to Q 0/1-Q 7/1, Q 8/1-Q 15/1 tPLH	-	-	-	71	56	71	nS	11a
12	33-48	CLK 1/2, CLK 2/2 to Q 0/2-Q 7/2, Q 8/2-Q 15/2 tPHL	-	-	-	73	58	73	nS	11a
12	49-64	CLK 1/2, CLK 2/2 to Q 0/2-Q 7/2, Q 8/2-Q 15/2 tPLH	-	-	-	71	56	71	nS	11a
14	65-96	QOE 1/1, QOE 4/1, QOE 1/2, QOE 4/2 to Q 0/1-Q 7/1, Q 8/1-Q 15/1, Q 0/2-Q 7/2, Q 8/2-Q 15/2 tPZH	-	-	-	69	53	69	nS	11b
14	97-128	QOE 1/1, QOE 4/1, QOE 1/2, QOE 4/2 to Q 0/1-Q 7/1, Q 8/1-Q 15/1, Q 0/2-Q 7/2, Q 8/2-Q 15/2 tPZL	-	-	-	71	55	71	nS	11b
17	129-192	-QOE 2/1, -QOE 3/1, -QOE 5/1, -QOE 6/1, -QOE 2/2, -QOE 3/2, -QOE 5/2, -QOE 6/2 to Q 0/1-Q 7/1, Q 8/1-Q 15/1, Q 0/2-Q 7/2, Q 8/2-Q 15/2 tPZH	-	-	-	99	75	99	nS	11b
17	193-256	-QOE 2/1, -QOE 3/1, -QOE 5/1, -QOE 6/1, -QOE 2/2, -QOE 3/2, -QOE 5/2, -QOE 6/2 to Q 0/1-Q 7/1, Q 8/1-Q 15/1, Q 0/2-Q 7/2, Q 8/2-Q 15/2 tPZL	-	-	-	101	77	101	nS	11b

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
15	257	QOE 4/2 to -QOE tPHL	-	-	-	35	28	35	nS	11d
16	258	QOE 4/2 to -QOE tPLH	-	-	-	34	27	34	nS	11d
19	259-260	-QOE 5/2,-QOE 6/2 to -QOE tPHL	-	-	-	65	50	65	nS	11d
20	261-262	-QOE 5/2,-QOE 6/2 to -QOE tPLH	-	-	-	62	43	62	nS	11d
18	263-294	DOE 1/1, DOE 1/2 to D 0/1-D 15/1, D 0/2-D 15/2 tPZH	-	-	-	69	53	69	nS	11c
18	295-326	DOE 1/1, DOE 1/2 to D 0/1-D 15/1, D 0/2-D 15/2 tPZL	-	-	-	71	55	71	nS	11c
21	327-390	-DOE 2/1, -DOE 3/1, -DOE 2/2, -DOE 3/2 to D 0/1-D 15/1, D 0/2-D 15/2 tPZH	-	-	-	99	75	99	nS	11c
21	391-454	-DOE 2/1, -DOE 3/1, -DOE 2/2, -DOE 3/2 to D 0/1-D 15/1, D 0/2-D 15/2 tPZL	-	-	-	101	77	101	nS	11c
2	455-458	CLK 1/1, CLK 2/1, CLK 1/2, CLK 2/2 (pulse width high) PWH	20	20	20	-	-	-	nS	11a
3	459-462	CLK 1/1, CLK 2/1, CLK 1/2, CLK 2/2 (pulse width low) PWL	20	20	20	-	-	-	nS	11a

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
1	463	CLK 1/1, CLK 2/1, CLK 1/2, CLK 2/2 (maximum frequency) Fmax	-	-	-	12	12	12	MHz	11a
6	464-467	LE 1/1, LE 1/2 to CLK 1/1, CLK 2/1, CLK 1/2, CLK 2/2, HL (setup time) tS	49	49	49	-	-	-	nS	11a
8	468-475	-LE 2/1, -LE 3/1, -LE 3/2, -LE 2/2 to CLK 1/1, CLK 2/1, CLK 1/2, CLK 2/2, HL (setup time) tS	79	79	79	-	-	-	nS	11a
11	476-507	D 0/1-D 7/1, D 8/1-D 15/1 D 0/2-D 7/2, D 8/2-D 15/2 to CLK 1/1, CLK 2/1, CLK 1/2, CLK 2/2 (setup time) tS	20	20	20	-	-	-	nS	11a
4	508-511	CLK 1/1, CLK 2/1, CLK 1/2, CLK 2/2 to LE 1/1, LE 1/2 (hold time) tH	3	3	3	-	-	-	nS	11a
7	512-519	CLK 1/1, CLK 2/1, CLK 1/2, CLK 2/2 to -LE 2/1, -LE 3/1, -LE 2/2, -LE 3/2 (hold time) tH	1	1	1	-	-	-	nS	11a
10	520-551	CLK 1/1, CLK 2/1, CLK 1/2, CLK 2/2 to D 0/1-D 7/1, D 8/1-D 15/1, D 0/2-D 7/2, D 8/2-D 15/2 (hold time) tH	5	5	5	-	-	-	nS	11a

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
5	552-555	LE 1/1, LE 1/2 to CLK 1/1 CLK 2/1, CLK 1/2, CLK 2/2 (setup time) tS	34	34	34	-	-	-	nS	11a
9	556-563	-LE 2/1, -LE 3/1, -LE 2/2, -LE 3/2 to CLK 1/1, CLK 2/1, CLK 1/2, CLK 2/2 (setup time) tS	64	64	64	-	-	-	nS	11a
12	564-579	CLK 1/1, CLK 2/1 to Q 0/1-Q 7/1, Q 8/1-Q 15/1 tPHL	-	-	-	84	69	84	nS	11a
12	580-595	CLK 1/1, CLK 2/1 to Q 0/1-Q 7/1, Q 8/1-Q 15/1 tPLH	-	-	-	80	65	80	nS	11a
12	596-611	CLK 1/2, CLK 2/2 to Q 0/2-Q 7/2, Q 8/2-Q 15/2 tPHL	-	-	-	84	69	84	nS	11a
12	612-627	CLK 1/2, CLK 2/2 to Q 0/2-Q 7/2, Q 8/2-Q 15/2 tPLH	-	-	-	80	65	80	nS	11a
14	628-659	QOE 1/1, QOE 4/1, QOE 1/2, QOE 4/2 to Q 0/1-Q 7/1, Q 8/1-Q 15/1, Q 0/2-Q 7/2 Q 8/2-Q 15/2 tPZH	-	-	-	78	62	78	nS	11b
14	660-691	QOE 1/1, QOE 4/1, QOE 1/2, QOE 4/2 to Q 0/1-Q 7/1, Q 8/1-Q 15/1, Q 0/2-Q 7/2, Q 8/2-Q 15/2 tPZL	-	-	-	82	66	82	nS	11b
17	692-755	-QOE 2/1, -QOE 3/1, -QOE 5/1, -QOE 6/1, -QOE 2/2, -QOE 3/2, -QOE 5/2, -QOE 6/2 to Q 0/1-Q 7/1, Q 8/1-Q 15/1, Q 0/2-Q 7/2, Q 8/2-Q 15/2 tPZH	-	-	-	108	84	108	nS	11b

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
17	756-819	-QOE 2/1, -QOE 3/1, -QOE 5/1, -QOE 6/1, -QOE 2/2, -QOE 3/2, -QOE 5/2, -QOE 6/2 to Q 0/1-Q 7/1, Q 8/1-Q 15/1 Q 0/2-Q 7/2, Q 8/2-Q 15/2 tPZL	-	-	-	112	88	112	nS	11b
15	820	QOE 4/2 to -QOE tPHL	-	-	-	51	44	51	nS	11d
16	821	QOE 4/2 to -QOE tPLH	-	-	-	50	45	50	nS	11d
19	822-823	-QOE 5/2, -QOE 6/2 to -QOE tPHL	-	-	-	81	66	81	nS	11d
20	824-825	-QOE 5/2, -QOE 6/2 to -QOE tPLH	-	-	-	80	61	80	nS	11d
18	826-857	DOE 1/1, DOE 1/2 to D 0/1-D 15/1, D 0/2-D15/2 tPZH	-	-	-	78	62	78	nS	11c
18	858-889	DOE 1/1, DOE 1/2 to D 0/1-D 15/1, D 0/2-D15/2 tPZL	-	-	-	82	66	82	nS	11c
21	890-953	-DOE 2/1, -DOE 3/1, -DOE 2/2, -DOE 3/2 to D 0/1-D 15/1, D 0/2-D 15/2 tPZH	-	-	-	108	84	108	nS	11c
21	954-1017	-DOE 2/1, -DOE 3/1, -DOE 2/2, -DOE 3/2 to D 0/1-D 15/1, D 0/2-D 15/2 tPZL	-	-	-	112	88	112	nS	11c

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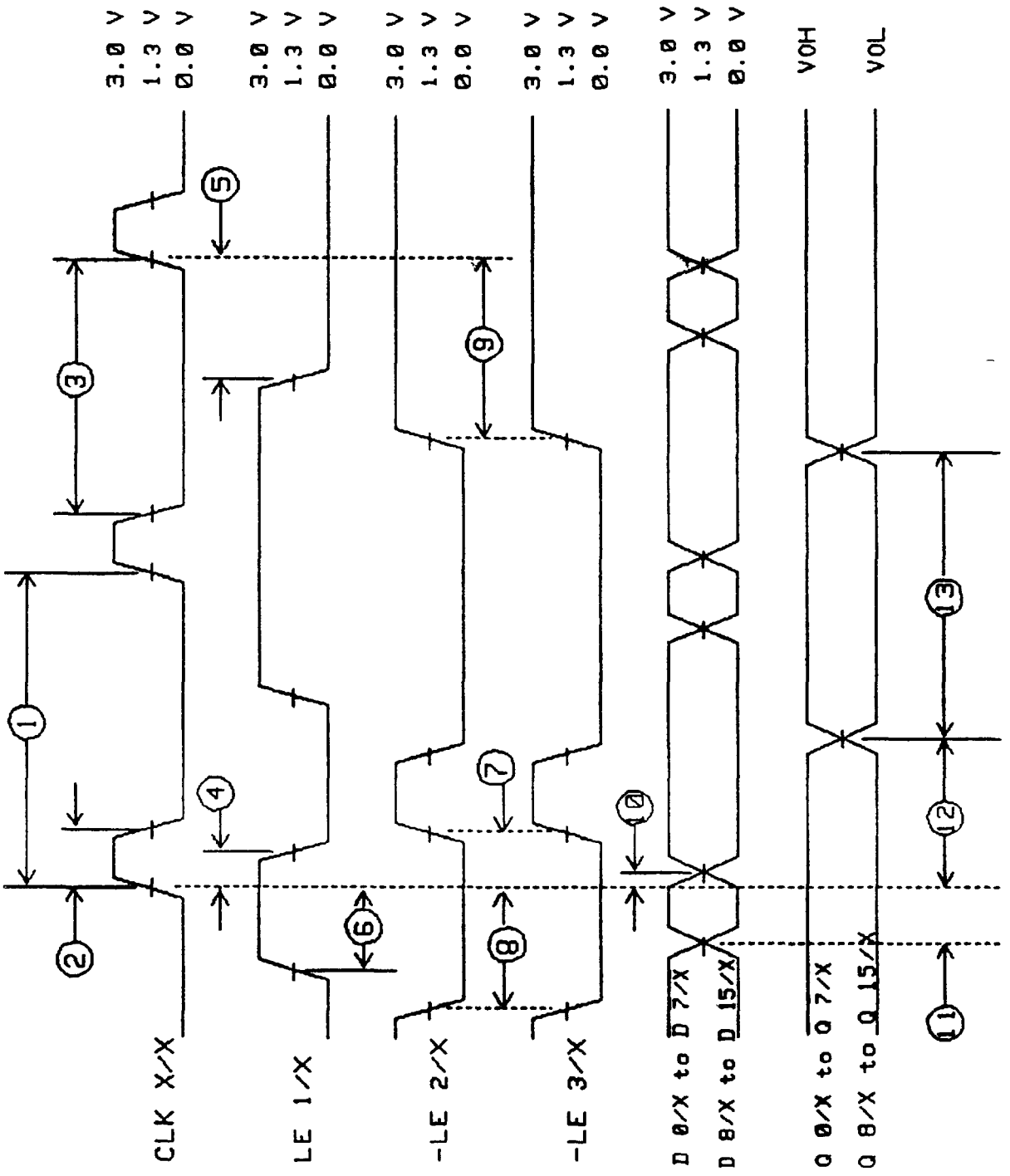


FIGURE 11a. Dynamic waveforms for table XVII.

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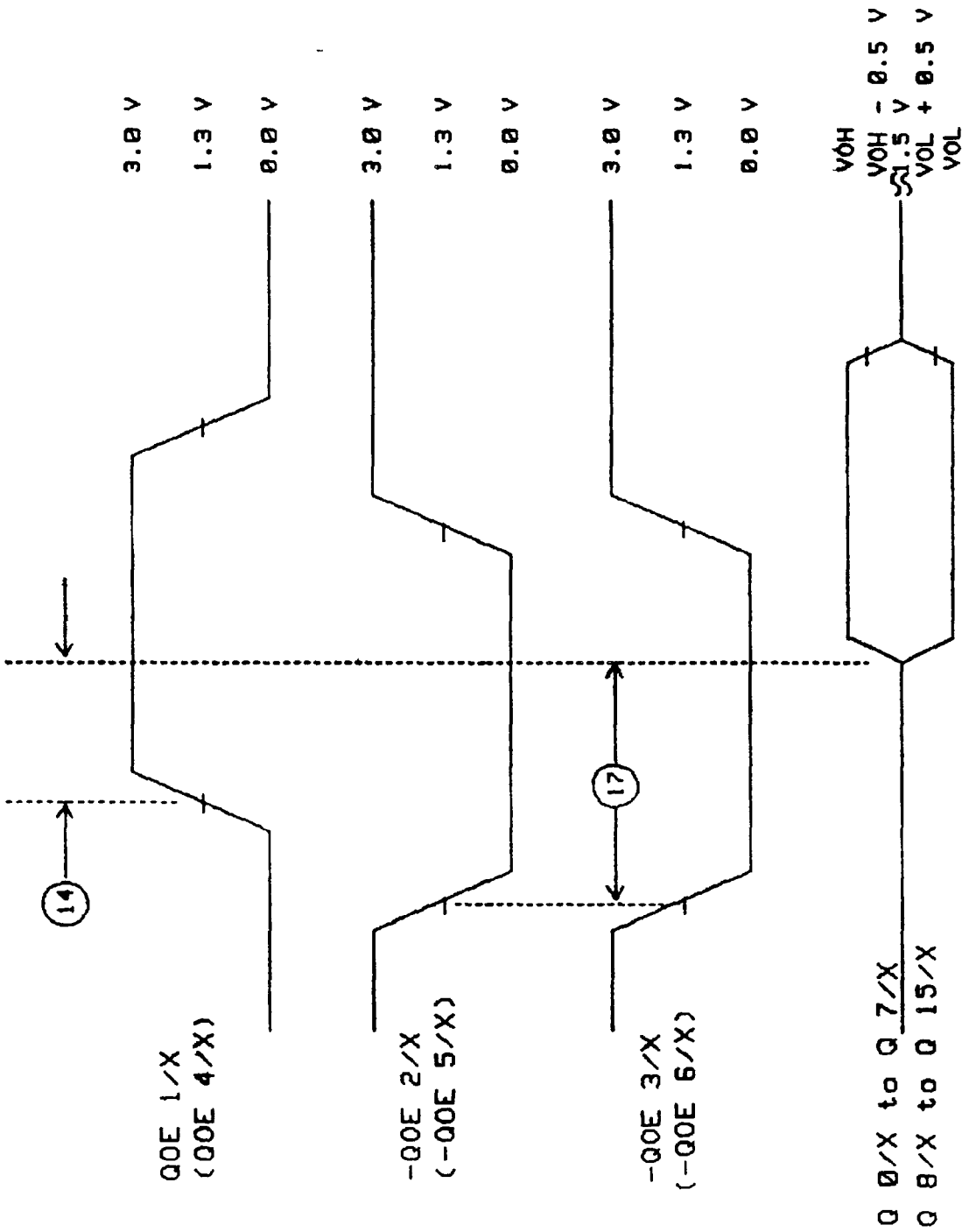


FIGURE 11b. Dynamic waveforms for table XVII.

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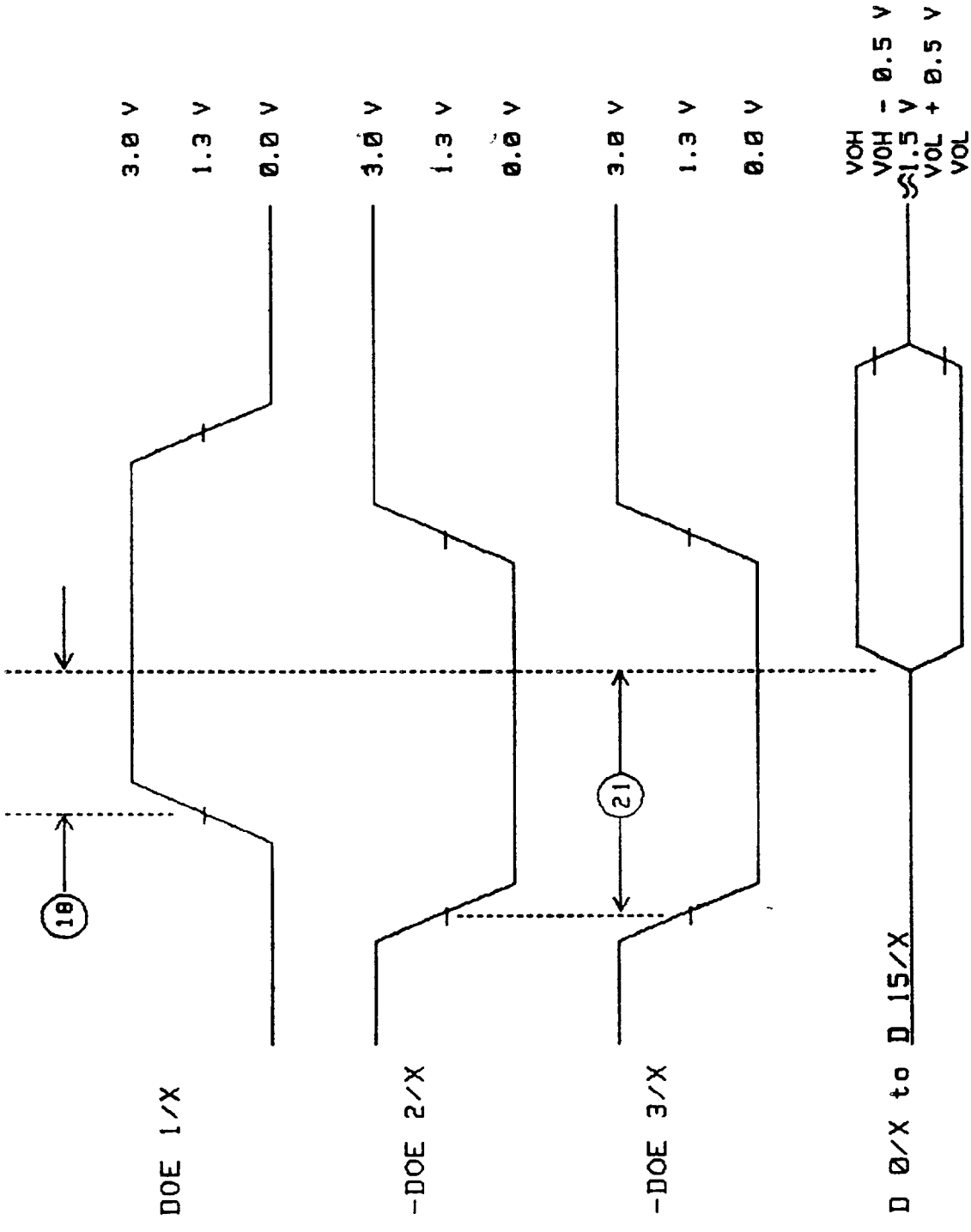


FIGURE 11c. Dynamic waveforms for table XVII.

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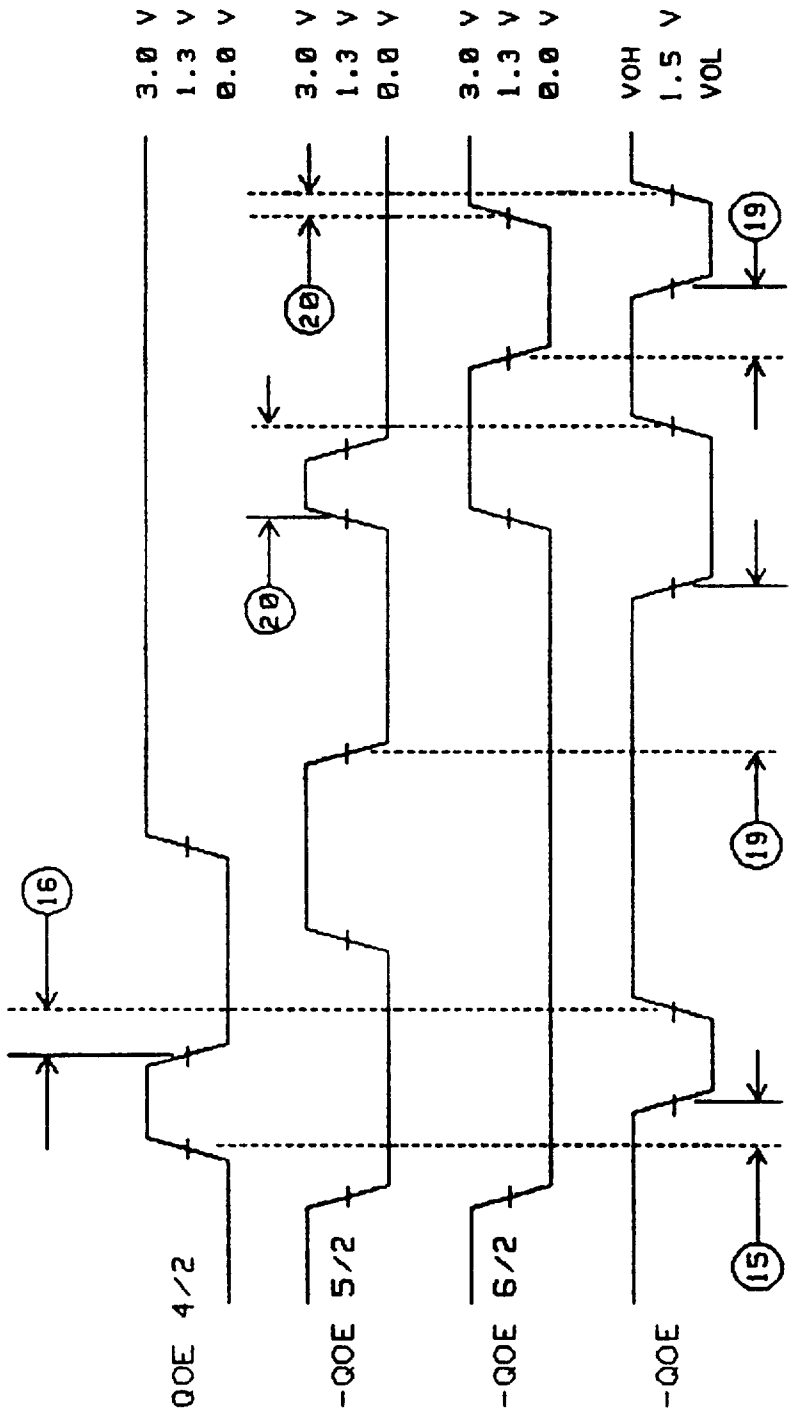


FIGURE 11d. Dynamic waveforms for table XVII.

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TABLE XVIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
R	PSFC	1,31	4.0	7.4	uF	1-3

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	R
Capacitance bridge voltage	0.3 Vp	± 30 mVp	R
Capacitance measurement	NA	± 170 nF	R

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. Abbreviations used are defined in 6.1.

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TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
S	PSTA	1	0	400	mV	1-4

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	S
Power supply current limit	0.878 A	NA	S
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 S S
Power supply transient amplitude measurement	NA	± 20 mV	S

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XIX.

(c) All Vcc pins shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XIX.

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NOTES: - Continued.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 12 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 12 pF. There shall be no other capacitance present on Vcc to ground.

3. Module inputs shall not violate the minimum input timing parameters referenced on figure 9.

4. Abbreviations used are defined in 6.1.

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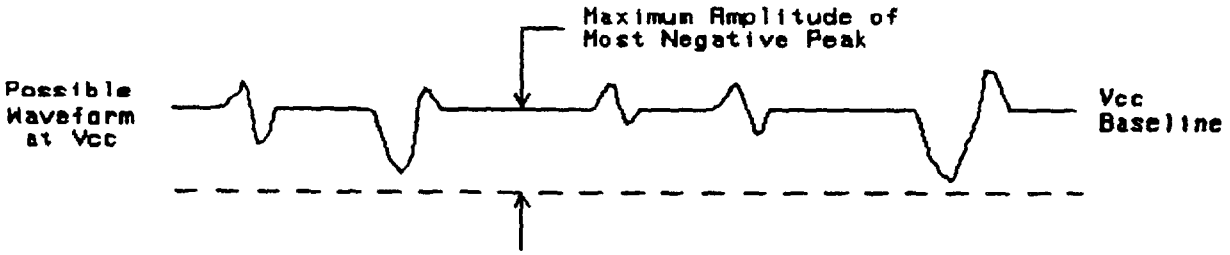


FIGURE 12. Power supply transient amplitude waveform for table XIX.

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TABLE XX. Output crosstalk, high level, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
T1	Q 0/1	63	0	500	mV	1-4	13,14
T2	Q 1/1	13					
T3	Q 2/1	62					
T4	Q 3/1	12					
T5	Q 4/1	7					
T6	Q 5/1	56					
T7	Q 6/1	6					
T8	Q 7/1	55					
T9	Q 8/1	17					
T10	Q 9/1	18					
T11	Q 10/1	68					
T12	Q 11/1	67					
T13	Q 12/1	64					
T14	Q 13/1	14					
T15	Q 14/1	16					
T16	Q 15/1	66					
T17	Q 0/2	87					
T18	Q 1/2	37					
T19	Q 2/2	86					
T20	Q 3/2	35	0	500	mV	1-4	13,14

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TABLE XX. Output crosstalk, high level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
T21	Q 4/2	82	0	500	mV	1-4	13,14
T22	Q 5/2	32					
T23	Q 6/2	81					
T24	Q 7/2	34					
T25	Q 8/2	39					
T26	Q 9/2	89					
T27	Q 10/2	93					
T28	Q 11/2	90					
T29	Q 12/2	38					
T30	Q 13/2	88					
T31	Q 14/2	36					
T32	Q 15/2	85					
T33	D 0/1	57					
T34	D 1/1	8					
T35	D 2/1	58					
T36	D 3/1	9					
T37	D 4/1	59					
T38	D 5/1	60					
T39	D 6/1	11					
T40	D 7/1	61	0	500	mV	1-4	13,14

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TABLE XX. Output crosstalk, high level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
T41	D 8/1	73	0	500	mV	1-4	13,14
T42	D 9/1	22					
T43	D 10/1	71					
T44	D 11/1	21					
T45	D 12/1	23					
T46	D 13/1	24					
T47	D 14/1	74					
T48	D 15/1	75					
T49	D 0/2	30					
T50	D 1/2	78					
T51	D 2/2	28					
T52	D 3/2	77					
T53	D 4/2	27					
T54	D 5/2	26					
T55	D 6/2	76					
T56	D 7/2	29					
T57	D 8/2	91					
T58	D 9/2	41					
T59	D 10/2	92					
T60	D 11/2	42	0	500	mV	1-4	13,14

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TABLE XX. Output crosstalk, high level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Initial limits Max	Units	Notes	Figs
T61	D 12/2	95	0	500	mV	1-4	13,14
T62	D 13/2	43					
T63	D 14/2	94					
T64	D 15/2	44					
T65	-QOE	100					
T66	PU	51	0	500	mV	1-4	13,14

NOTES:

1. Output crosstalk, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 0.25 V	T1-T66
Power supply current limit	0.878 A	NA	T1-T66
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T1-T66 T1-T66
Output crosstalk measurement	NA	± 25 mV	T1-T66

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line. Module inputs shall not violate the minimum input timing parameters referenced on figure 9.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS $\pm$ 200 nS, 50 percent duty cycle.

(d) For each output under test in table XX, connect probe 'B' and visually inspect the output for the most negative spike as indicated on figure 13.

(e) The maximum probe capacitance is 10 pF.

3. The test circuit shall be as indicated on figure 14. All outputs shall be loaded as specified on figure 8.

4. Abbreviations used are defined in 6.1.

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TABLE XXI. Output crosstalk, low level, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs
U1	Q 0/1	63	0	900	mV	1-4	13,14
U2	Q 1/1	13					
U3	Q 2/1	62					
U4	Q 3/1	12					
U5	Q 4/1	7					
U6	Q 5/1	56					
U7	Q 6/1	6					
U8	Q 7/1	55					
U9	Q 8/1	17					
U10	Q 9/1	18					
U11	Q 10/1	68					
U12	Q 11/1	67					
U13	Q 12/1	64					
U14	Q 13/1	14					
U15	Q 14/1	16					
U16	Q 15/1	66					
U17	Q 0/2	87					
U18	Q 1/2	37					
U19	Q 2/2	86					
U20	Q 3/2	35	0	900	mV	1-4	13,14

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TABLE XXI. Output crosstalk, low level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
U21	Q 4/2	82	0	900	mV	1-4	13,14
U22	Q 5/2	32					
U23	Q 6/2	81					
U24	Q 7/2	34					
U25	Q 8/2	39					
U26	Q 9/2	89					
U27	Q 10/2	93					
U28	Q 11/2	90					
U29	Q 12/2	38					
U30	Q 13/2	88					
U31	Q 14/2	36					
U32	Q 15/2	85	0	900	mV	1-4	13,14
U33	D 0/1	57	0	500	mV	1-4	13,14
U34	D 1/1	8					
U35	D 2/1	58					
U36	D 3/1	9					
U37	D 4/1	59					
U38	D 5/1	60					
U39	D 6/1	11					
U40	D 7/1	61	0	500	mV	1-4	13,14

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TABLE XXI. Output crosstalk, low level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs
U41	D 8/1	73	0	500	mV	1-4	13,14
U42	D 9/1	22					
U43	D 10/1	71					
U44	D 11/1	21					
U45	D 12/1	23					
U46	D 13/1	24					
U47	D 14/1	74					
U48	D 15/1	75					
U49	D 0/2	30					
U50	D 1/2	78					
U51	D 2/2	28					
U52	D 3/2	77					
U53	D 4/2	27					
U54	D 5/2	26					
U55	D 6/2	76					
U56	D 7/2	29					
U57	D 8/2	91					
U58	D 9/2	41					
U59	D 10/2	92					
U60	D 11/2	42	0	500	mV	1-4	13,14

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TABLE XXI. Output crosstalk, low level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs
U61	D 12/2	95	0	500	mV	1-4	13,14
U62	D 13/2	43					
U63	D 14/2	94					
U64	D 15/2	44					
U65	-QOE	100	0	500	mV	1-4	13,14

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NOTES:

1. Output crosstalk, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 0.25 V	U1-U65
Power supply current limit	0.878 A	NA	U1-U65
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 U1-U65 U1-U65
Output crosstalk measurement	 NA NA	 ± 45 mV ± 25 mV	 U1-U32 U33-U65

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line. Module inputs shall not violate the minimum input timing parameters referenced on figure 9.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

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NOTES: - Continued.

(d) For each output under test in table XXI, connect probe 'B' and visually inspect the output for the most positive spike as indicated on figure 13.

(e) The maximum probe capacitance is 10 pF.

3. The test circuit shall be as indicated on figure 14. All outputs shall be loaded as specified on figure 8.

4. Abbreviations used are defined in 6.1.

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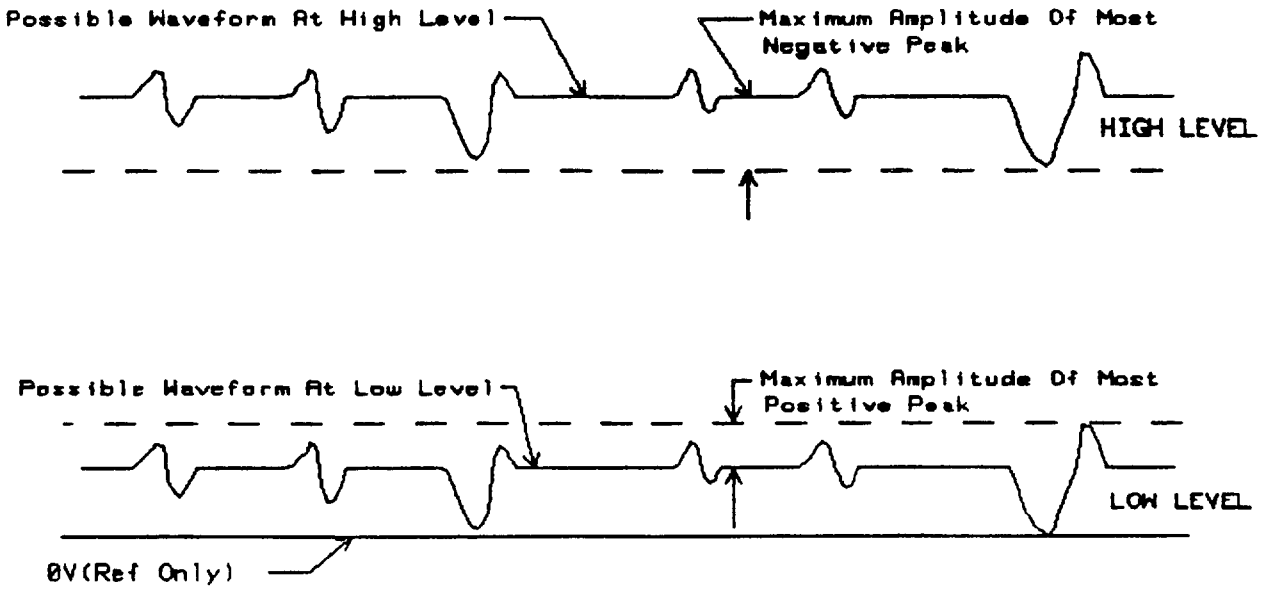


FIGURE 13. Output crosstalk, high and low level, test waveforms for tables XX and XXI.

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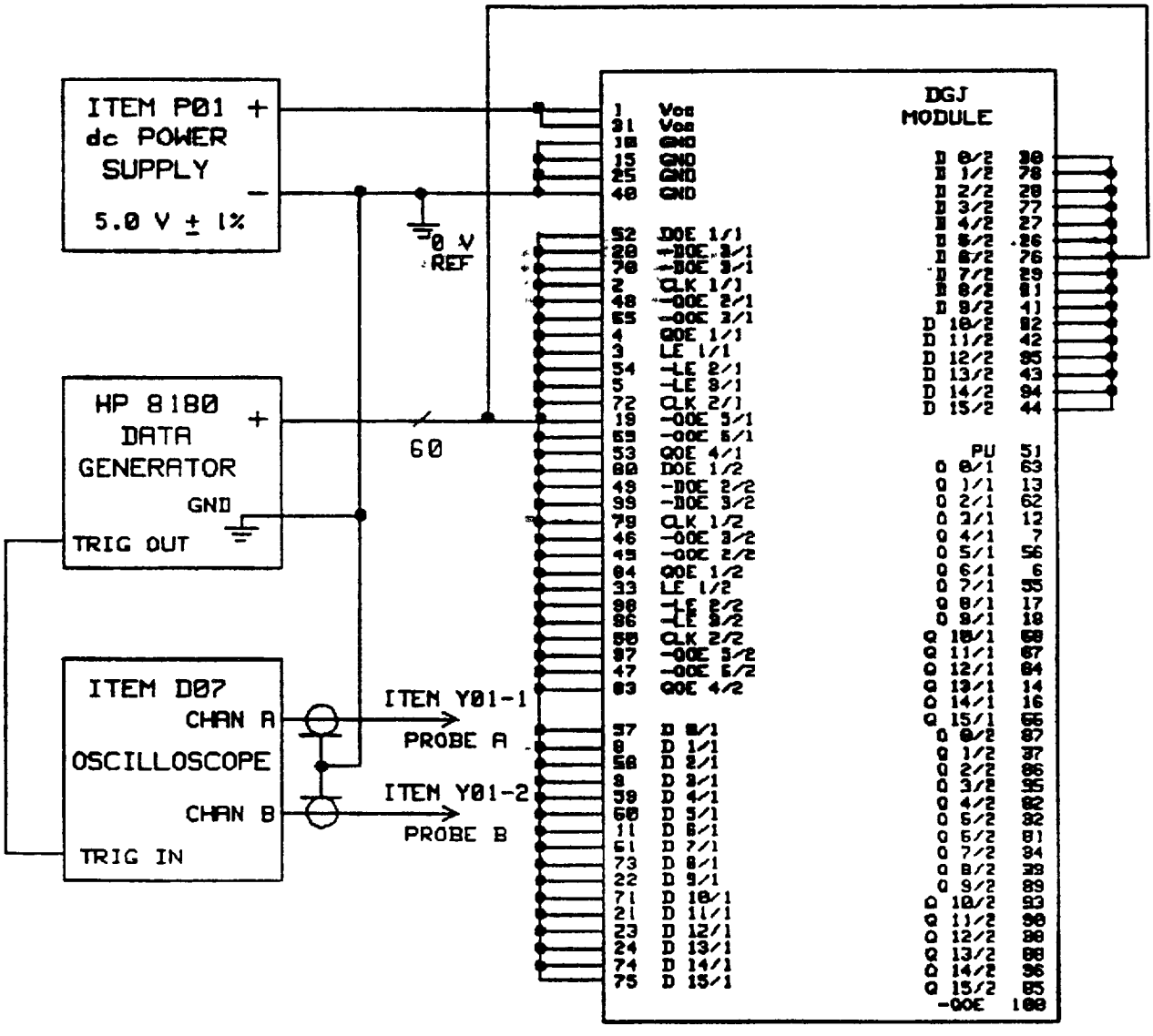


FIGURE 14. Output crosstalk, high and low level, test circuit for tables XX, XXI, and XXII.

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TABLE XXII. Output crosstalk duration test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Fig
V1	Q 0/1	63	0	70	nS	1-4	14,15
V2	Q 1/1	13					
V3	Q 2/1	62					
V4	Q 3/1	12					
V5	Q 4/1	7					
V6	Q 5/1	56					
V7	Q 6/1	6					
V8	Q 7/1	55					
V9	Q 8/1	17					
V10	Q 9/1	18					
V11	Q 10/1	68					
V12	Q 11/1	67					
V13	Q 12/1	64					
V14	Q 13/1	14					
V15	Q 14/1	16					
V16	Q 15/1	66					
V17	Q 0/2	87					
V18	Q 1/2	37					
V19	Q 2/2	86					
V20	Q 3/2	35	0	70	nS	1-4	14,15

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TABLE XXII. Output crosstalk duration test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Fig
V21	Q 4/2	82	0	70	nS	1-4	14,15
V22	Q 5/2	32					
V23	Q 6/2	81					
V24	Q 7/2	34					
V25	Q 8/2	39					
V26	Q 9/2	89					
V27	Q 10/2	93					
V28	Q 11/2	90					
V29	Q 12/2	38					
V30	Q 13/2	88					
V31	Q 14/2	36					
V32	Q 15/2	85	0	70	nS	1-4	14,15

NOTES:

1. Output crosstalk duration test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	V1-V32
Power supply current limit	0.878 A	NA	V1-V32

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 V1-V32 V1-V32
Crosstalk duration measurement	NA	± 3.5 nS	V1-V32

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

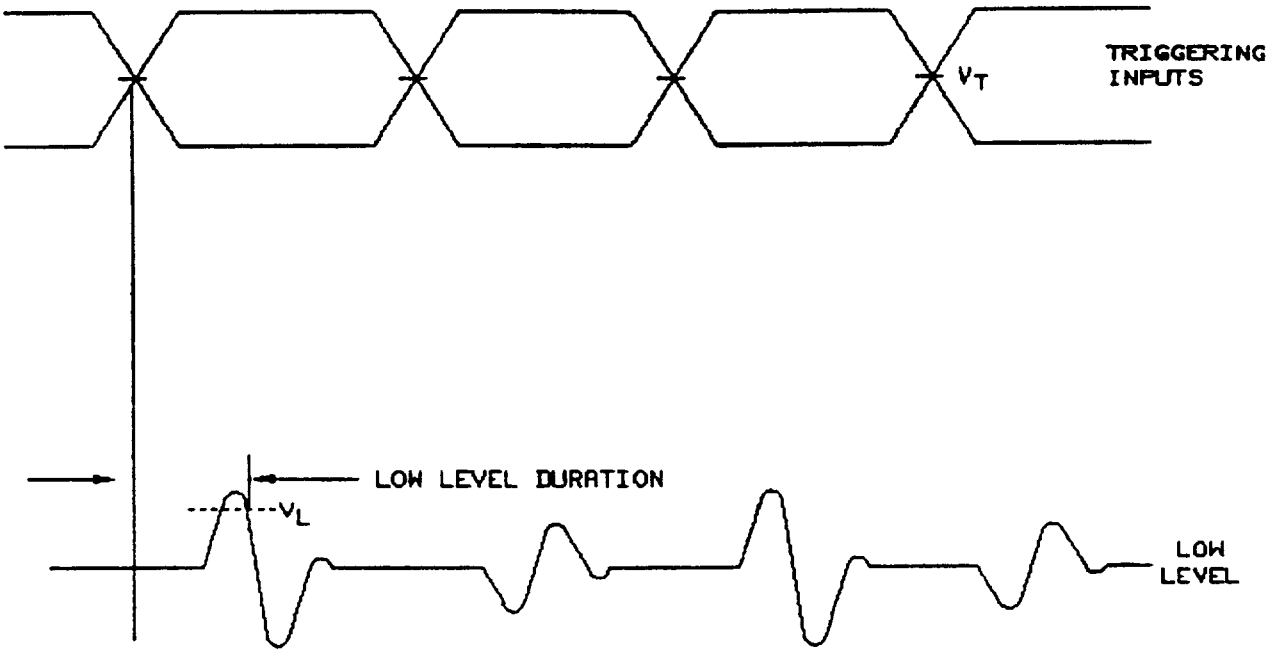
(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line. Module inputs shall not violate the minimum input timing parameters referenced on figure 9.

(c) The duration shall be measured from the triggering edge of the input as shown on figure 15.

3. All outputs shall be loaded as specified on figure 8.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. VT = 1.3 V, VL = 400 mV (AC coupled).

2. The crosstalk duration is measured from the triggering edge of the input to the point where output crosstalk returns to the specified voltage. If the crosstalk amplitude is smaller than the duration measurement voltage (VL), the crosstalk duration shall be recorded as 0 nS.

FIGURE 15. Output crosstalk duration test waveforms for table XXII.

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TABLE XXIII. Input capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
W1	CLK 1/1	2	2	30	pF	1-3
W2	-DOE 2/1	20				
W3	-QOE 6/2	47				
W4	DOE 1/1	52				
W5	CLK 2/2	50				
W6	-DOE 2/2	49				
W7	-DOE 3/1	70				
W8	CLK 2/1	72				
W9	CLK 1/2	79				
W10	DOE 1/2	80				
W11	QOE 4/2	83				
W12	QOE 1/2	84				
W13	-QOE 5/2	97				
W14	-DOE 3/2	99				
W15	QOE 1/1	4				
W16	-QOE 5/1	19				
W17	-QOE 3/2	46				
W18	-QOE 3/1	65				
W19	-QOE 6/1	69				
W20	LE 1/1	3	2	30	pF	1-3

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TABLE XXIII. Input capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
W21	-LE 3/1	5	2	30	pF	1-3
W22	LE 1/2	33				
W23	-QOE 2/2	45				
W24	QOE 4/1	53				
W25	-LE 2/1	54				
W26	-LE 3/2	96				
W27	-LE 2/2	98	2	30	pF	1-3
W28	-QOE 2/1	48	2	50	pF	1-3

NOTES:

1. Input capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	W1-W28
Capacitance meter voltage	15 mV	± 1.5 mV	W1-W28
Capacitance measurement	NA NA	± 1.4 pF ± 2.4 pF	W1-W27 W28

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XXIII.

3. Abbreviations used are defined in 6.1.

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TABLE XXIV. Output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
X1	Q 6/1	6	2	40	pF	1-3
X2	Q 4/1	7				
X3	Q 3/1	12				
X4	Q 1/1	13				
X5	Q 13/1	14				
X6	Q 14/1	16				
X7	Q 8/1	17				
X8	Q 9/1	18				
X9	Q 5/2	32				
X10	Q 7/2	34				
X11	Q 3/2	35				
X12	Q 14/2	36				
X13	Q 1/2	37				
X14	Q 12/2	38				
X15	Q 8/2	39				
X16	Q 7/1	55				
X17	Q 5/1	56				
X18	Q 2/1	62				
X19	Q 0/1	63				
X20	Q 12/1	64	2	40	pF	1-3

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TABLE XXIV. Output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
X21	Q 15/1	66	2	40	pF	1-3
X22	Q 11/1	67				
X23	Q 10/1	68				
X24	Q 6/2	81				
X25	Q 4/2	82				
X26	Q 15/2	85				
X27	Q 2/2	86				
X28	Q 0/2	87				
X29	Q 13/2	88				
X30	Q 9/2	89				
X31	Q 11/2	90				
X32	Q 10/2	93	2	40	pF	1-3
X33	D 1/1	8	2	45	pF	1-3
X34	D 3/1	9				
X35	D 6/1	11				
X36	D 11/1	21				
X37	D 9/1	22				
X38	D 12/1	23				
X39	D 13/1	24				
X40	D 5/2	26	2	45	pF	1-3

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TABLE XXIV. Output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
X41	D 4/2	27	2	45	pF	1-3
X42	D 2/2	28				
X43	D 7/2	29				
X44	D 0/2	30				
X45	D 9/2	41				
X46	D 11/2	42				
X47	D 13/2	43				
X48	D 15/2	44				
X49	D 0/1	57				
X50	D 2/1	58				
X51	D 4/1	59				
X52	D 5/1	60				
X53	D 7/1	61				
X54	D 10/1	71				
X55	D 8/1	73				
X56	D 14/1	74				
X57	D 15/1	75				
X58	D 6/2	76				
X59	D 3/2	77				
X60	D 1/2	78	2	45	pF	1-3

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TABLE XXIV. Output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
X61	D 8/2	91	2	45	pF	1-3
X62	D 10/2	92				
X63	D 14/2	94				
X64	D 12/2	95	2	45	pF	1-3

NOTES:

1. Output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	X1-X64
Capacitance meter voltage	15 mV	± 1.5 mV	X1-X64
Capacitance measurement	NA NA	± 1.9 pF ± 2.15 pF	X1-X32 X33-X64

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XXIV.

3. Abbreviations used are defined in 6.1.

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TABLE XXV. Threshold test.

Test cond	Parameter	Notes	Fig
Y	Threshold	1-4	8

NOTES:

1. Threshold test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	Y
Power supply current limit	0.878 A	NA	Y
Pattern voltage high, (VH) low, (VL)	 2.0 V 0.7 V	 ± 20 mV ± 7 mV	 Y Y
Output voltage measurement high, (Voh) low, (Vol)	 NA NA	 ± 105 mV ± 20 mV	 Y Y

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and verify the output patterns by monitoring the outputs and comparing the results to the output low and high levels specified in the output voltage tests. The module inputs shall not violate the minimum input timing parameters referenced on figure 9.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

3. The load for each output is specified on figure 8.

4. Abbreviations used are defined in 6.1.

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TABLE XXVI. Disable time test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
Low level to high impedance(tPLZ)						
Z1	QOE 1/1 to Q 0/1	63	4	45	nS	22
Z2	QOE 1/1 to Q 1/1	13				
Z3	QOE 1/1 to Q 2/1	62				
Z4	QOE 1/1 to Q 3/1	12				
Z5	QOE 1/1 to Q 4/1	7				
Z6	QOE 1/1 to Q 5/1	56				
Z7	QOE 1/1 to Q 6/1	6				
Z8	QOE 1/1 to Q 7/1	55				
Z9	QOE 4/1 to Q 8/1	17				
Z10	QOE 4/1 to Q 9/1	18				
Z11	QOE 4/1 to Q 10/1	68				
Z12	QOE 4/1 to Q 11/1	67				
Z13	QOE 4/1 to Q 12/1	64				
Z14	QOE 4/1 to Q 13/1	14				
Z15	QOE 4/1 to Q 14/1	16				
Z16	QOE 4/1 to Q 15/1	66				
Z17	QOE 1/2 to Q 0/2	87				
Z18	QOE 1/2 to Q 1/2	37				
Z19	QOE 1/2 to Q 2/2	86				
Z20	QOE 1/2 to Q 3/2	35	4	45	nS	22

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TABLE XXVI. Disable time test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
Z21	QOE 1/2 to Q 4/2	82	4	45	nS	22
Z22	QOE 1/2 to Q 5/2	32				
Z23	QOE 1/2 to Q 6/2	81				
Z24	QOE 1/2 to Q 7/2	34				
Z25	QOE 4/2 to Q 8/2	39				
Z26	QOE 4/2 to Q 9/2	89				
Z27	QOE 4/2 to Q 10/2	93				
Z28	QOE 4/2 to Q 11/2	90				
Z29	QOE 4/2 to Q 12/2	38				
Z30	QOE 4/2 to Q 13/2	88				
Z31	QOE 4/2 to Q 14/2	36				
Z32	QOE 4/2 to Q 15/2	85	4	45	nS	22
High level to high impedance(tPHZ)						
Z33	QOE 1/1 to Q 0/1	63	4	50	nS	22
Z34	QOE 1/1 to Q 1/1	13				
Z35	QOE 1/1 to Q 2/1	62				
Z36	QOE 1/1 to Q 3/1	12				
Z37	QOE 1/1 to Q 4/1	7				
Z38	QOE 1/1 to Q 5/1	56				
Z39	QOE 1/1 to Q 6/1	6				
Z40	QOE 1/1 to Q 7/1	55	4	50	nS	22

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TABLE XXVI. Disable time test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
Z41	QOE 4/1 to Q 8/1	17	4	50	nS	22
Z42	QOE 4/1 to Q 9/1	18				
Z43	QOE 4/1 to Q 10/1	68				
Z44	QOE 4/1 to Q 11/1	67				
Z45	QOE 4/1 to Q 12/1	64				
Z46	QOE 4/1 to Q 13/1	14				
Z47	QOE 4/1 to Q 14/1	16				
Z48	QOE 4/1 to Q 15/1	66				
Z49	QOE 1/2 to Q 0/2	87				
Z50	QOE 1/2 to Q 1/2	37				
Z51	QOE 1/2 to Q 2/2	86				
Z52	QOE 1/2 to Q 3/2	35				
Z53	QOE 1/2 to Q 4/2	82				
Z54	QOE 1/2 to Q 5/2	32				
Z55	QOE 1/2 to Q 6/2	81				
Z56	QOE 4/2 to Q 7/2	34				
Z57	QOE 4/2 to Q 8/2	39				
Z58	QOE 4/2 to Q 9/2	89				
Z59	QOE 4/2 to Q 10/2	93				
Z60	QOE 4/2 to Q 11/2	90				
Z61	QOE 4/2 to Q 12/2	38	4	50	nS	22

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TABLE XXVI. Disable time test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Fig
Z62	QOE 4/2 to Q 13/2	88	4	50	nS	22
Z63	QOE 4/2 to Q 14/2	36				
Z64	QOE 4/2 to Q 15/2	85	4	50	nS	22
Low level to high impedance (tPLZ)						
Z65	-QOE 2/1 to Q 0/1	63	6	61	nS	23
Z66	-QOE 2/1 to Q 1/1	13				
Z67	-QOE 2/1 to Q 2/1	62				
Z68	-QOE 2/1 to Q 3/1	12				
Z69	-QOE 2/1 to Q 4/1	7				
Z70	-QOE 2/1 to Q 5/1	56				
Z71	-QOE 2/1 to Q 6/1	6				
Z72	-QOE 2/1 to Q 7/1	55				
Z73	-QOE 5/1 to Q 8/1	17				
Z74	-QOE 5/1 to Q 9/1	18				
Z75	-QOE 5/1 to Q 10/1	68				
Z76	-QOE 5/1 to Q 11/1	67				
Z77	-QOE 5/1 to Q 12/1	64				
Z78	-QOE 5/1 to Q 13/1	14				
Z79	-QOE 5/1 to Q 14/1	16				
Z80	-QOE 5/1 to Q 15/1	66				
Z81	-QOE 2/2 to Q 0/2	87	6	61	nS	23

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TABLE XXVI. Disable time test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Fig
Z82	-QOE 2/2 to Q 1/2	37	6	61	nS	23
Z83	-QOE 2/2 to Q 2/2	86				
Z84	-QOE 2/2 to Q 3/2	35				
Z85	-QOE 2/2 to Q 4/2	82				
Z86	-QOE 2/2 to Q 5/2	32				
Z87	-QOE 2/2 to Q 6/2	81				
Z88	-QOE 2/2 to Q 7/2	34				
Z89	-QOE 5/2 to Q 8/2	39				
Z90	-QOE 5/2 to Q 9/2	89				
Z91	-QOE 5/2 to Q 10/2	93				
Z92	-QOE 5/2 to Q 11/2	90				
Z93	-QOE 5/2 to Q 12/2	38				
Z94	-QOE 5/2 to Q 13/2	88				
Z95	-QOE 5/2 to Q 14/2	36				
Z96	-QOE 5/2 to Q 15/2	85	6	61	nS	23
High level to high impedance (tPHZ)						
Z97	-QOE 2/1 to Q 0/1	63	6	66	nS	23
Z98	-QOE 2/1 to Q 1/1	13				
Z99	-QOE 2/1 to Q 2/1	62				
Z100	-QOE 2/1 to Q 3/1	12				
Z101	-QOE 2/1 to Q 4/1	7	6	66	nS	23

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TABLE XXVI. Disable time test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
Z102	-QOE 2/1 to Q 5/1	56	6	66	nS	23
Z103	-QOE 2/1 to Q 6/1	6				
Z104	-QOE 2/1 to Q 7/1	55				
Z105	-QOE 5/1 to Q 8/1	17				
Z106	-QOE 5/1 to Q 9/1	18				
Z107	-QOE 5/1 to Q 10/1	68				
Z108	-QOE 5/1 to Q 11/1	67				
Z109	-QOE 5/1 to Q 12/1	64				
Z110	-QOE 5/1 to Q 13/1	14				
Z111	-QOE 5/1 to Q 14/1	16				
Z112	-QOE 5/1 to Q 15/1	66				
Z113	-QOE 2/2 to Q 0/2	87				
Z114	-QOE 2/2 to Q 1/2	37				
Z115	-QOE 2/2 to Q 2/2	86				
Z116	-QOE 2/2 to Q 3/2	35				
Z117	-QOE 2/2 to Q 4/2	82				
Z118	-QOE 2/2 to Q 5/2	32				
Z119	-QOE 2/2 to Q 6/2	81				
Z120	-QOE 2/2 to Q 7/2	34				
Z121	-QOE 5/2 to Q 8/2	39				
Z122	-QOE 5/2 to Q 9/2	89	6	66	nS	23

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TABLE XXVI. Disable time test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Fig
Z123	-QOE 5/2 to Q 10/2	93	6	66	nS	23
Z124	-QOE 5/2 to Q 11/2	90				
Z125	-QOE 5/2 to Q 12/2	38				
Z126	-QOE 5/2 to Q 13/2	88				
Z127	-QOE 5/2 to Q 14/2	36				
Z128	-QOE 5/2 to Q 15/2	85	6	66	nS	23
Low level to high impedance (tPLZ)						
Z129	-QOE 3/1 to Q 0/1	63	6	61	nS	23
Z130	-QOE 3/1 to Q 1/1	13				
Z131	-QOE 3/1 to Q 2/1	62				
Z132	-QOE 3/1 to Q 3/1	12				
Z133	-QOE 3/1 to Q 4/1	7				
Z134	-QOE 3/1 to Q 5/1	56				
Z135	-QOE 3/1 to Q 6/1	6				
Z136	-QOE 3/1 to Q 7/1	55				
Z137	-QOE 6/1 to Q 8/1	17				
Z138	-QOE 6/1 to Q 9/1	18				
Z139	-QOE 6/1 to Q 10/1	68				
Z140	-QOE 6/1 to Q 11/1	67				
Z141	-QOE 6/1 to Q 12/1	64				
Z142	-QOE 6/1 to Q 13/1	14	6	61	nS	23

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TABLE XXVI. Disable time test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Fig
Z143	-QOE 6/1 to Q 14/1	16	6	61	nS	23
Z144	-QOE 6/1 to Q 15/1	66				
Z145	-QOE 3/2 to Q 0/2	87				
Z146	-QOE 3/2 to Q 1/2	37				
Z147	-QOE 3/2 to Q 2/2	86				
Z148	-QOE 3/2 to Q 3/2	35				
Z149	-QOE 3/2 to Q 4/2	82				
Z150	-QOE 3/2 to Q 5/2	32				
Z151	-QOE 3/2 to Q 6/2	81				
Z152	-QOE 3/2 to Q 7/2	34				
Z153	-QOE 6/2 to Q 8/2	39				
Z154	-QOE 6/2 to Q 9/2	89				
Z155	-QOE 6/2 to Q 10/2	93				
Z156	-QOE 6/2 to Q 11/2	90				
Z157	-QOE 6/2 to Q 12/2	38				
Z158	-QOE 6/2 to Q 13/2	88				
Z159	-QOE 6/2 to Q 14/2	36				
Z160	-QOE 6/2 to Q 15/2	85	6	61	nS	23
High level to high impedance (tPHZ)						
Z161	-QOE 3/1 to Q 0/1	63	6	66	nS	23

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TABLE XXVI. Disable time test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Fig
Z162	-QOE 3/1 to Q 1/1	13	6	66	nS	23
Z163	-QOE 3/1 to Q 2/1	62				
Z164	-QOE 3/1 to Q 3/1	12				
Z165	-QOE 3/1 to Q 4/1	7				
Z166	-QOE 3/1 to Q 5/1	56				
Z167	-QOE 3/1 to Q 6/1	6				
Z168	-QOE 3/1 to Q 7/1	55				
Z169	-QOE 6/1 to Q 8/1	17				
Z170	-QOE 6/1 to Q 9/1	18				
Z171	-QOE 6/1 to Q 10/1	68				
Z172	-QOE 6/1 to Q 11/1	67				
Z173	-QOE 6/1 to Q 12/1	64				
Z174	-QOE 6/1 to Q 13/1	14				
Z175	-QOE 6/1 to Q 14/1	16				
Z176	1QOE 6/1 to Q 15/1	66				
Z177	-QOE 3/1 to Q 0/2	87				
Z178	-QOE 3/1 to Q 1/2	37				
Z179	-QOE 3/1 to Q 2/2	86				
Z180	-QOE 3/1 to Q 3/2	35				
Z181	-QOE 3/1 to Q 4/2	82				
Z182	-QOE 3/1 to Q 5/2	32	6	66	nS	23

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TABLE XXVI. Disable time test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Fig
Z183	-QOE 3/1 to Q 6/2	81	6	66	nS	23
Z184	-QOE 3/1 to Q 7/2	34				
Z185	-QOE 6/1 to Q 8/2	39				
Z186	-QOE 6/1 to Q 9/2	89				
Z187	-QOE 6/1 to Q 10/2	93				
Z188	-QOE 6/1 to Q 11/2	90				
Z189	-QOE 6/1 to Q 12/2	38				
Z190	-QOE 6/1 to Q 13/2	88				
Z191	-QOE 6/1 to Q 14/2	36				
Z192	-QOE 6/1 to Q 15/2	85	6	66	nS	23
Low level to high impedance (tPLZ)						
Z193	DOE 1/1 to D 0/1	57	4	45	nS	22
Z194	DOE 1/1 to D 1/1	8				
Z195	DOE 1/1 to D 2/1	58				
Z196	DOE 1/1 to D 3/1	9				
Z197	DOE 1/1 to D 4/1	59				
Z198	DOE 1/1 to D 5/1	60				
Z199	DOE 1/1 to D 6/1	11				
Z200	DOE 1/1 to D 7/1	61				
Z201	DOE 1/1 to D 8/1	73				
Z202	DOE 1/1 to D 9/1	22	4	45	nS	22

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TABLE XXVI. Disable time test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
Z203	DOE 1/1 to D 10/1	71	4	45	nS	22
Z204	DOE 1/1 to D 11/1	21				
Z205	DOE 1/1 to D 12/1	23				
Z206	DOE 1/1 to D 13/1	24				
Z207	DOE 1/1 to D 14/1	74				
Z208	DOE 1/1 to D 15/1	75				
Z209	DOE 1/2 to D 0/2	30				
Z210	DOE 1/2 to D 1/2	78				
Z211	DOE 1/2 to D 2/2	28				
Z212	DOE 1/2 to D 3/2	77				
Z213	DOE 1/2 to D 4/2	27				
Z214	DOE 1/2 to D 5/2	26				
Z215	DOE 1/2 to D 6/2	76				
Z216	DOE 1/2 to D 7/2	29				
Z217	DOE 1/2 to D 8/2	91				
Z218	DOE 1/2 to D 9/2	41				
Z219	DOE 1/2 to D 10/2	92				
Z220	DOE 1/2 to D 11/2	42				
Z221	DOE 1/2 to D 12/2	95				
Z222	DOE 1/2 to D 13/2	43				
Z223	DOE 1/2 to D 14/2	94	4	45	nS	22

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TABLE XXVI. Disable time test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Fig
Z224	DOE 1/2 to D 15/2	44	4	45	nS	22
High level to high impedance (tPHZ)						
Z225	DOE 1/1 to D 0/1	57	4	50	nS	22
Z226	DOE 1/1 to D 1/1	8				
Z227	DOE 1/1 to D 2/1	58				
Z228	DOE 1/1 to D 3/1	9				
Z229	DOE 1/1 to D 4/1	59				
Z230	DOE 1/1 to D 5/1	60				
Z231	DOE 1/1 to D 6/1	11				
Z232	DOE 1/1 to D 7/1	61				
Z233	DOE 1/1 to D 8/1	73				
Z234	DOE 1/1 to D 9/1	22				
Z235	DOE 1/1 to D 10/1	71				
Z236	DOE 1/1 to D 11/1	21				
Z237	DOE 1/1 to D 12/1	23				
Z238	DOE 1/1 to D 13/1	24				
Z239	DOE 1/1 to D 14/1	74				
Z240	DOE 1/1 to D 15/1	75				
Z241	DOE 1/2 to D 0/2	30				
Z242	DOE 1/2 to D 1/2	78				
Z243	DOE 1/2 to D 2/2	28	4	50	nS	22

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TABLE XXVI. Disable time test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
Z244	DOE 1/2 to D 3/2	77	4	50	nS	22
Z245	DOE 1/2 to D 4/2	27				
Z246	DOE 1/2 to D 5/2	26				
Z247	DOE 1/2 to D 6/2	76				
Z248	DOE 1/2 to D 7/2	29				
Z249	DOE 1/2 to D 8/2	91				
Z250	DOE 1/2 to D 9/2	41				
Z251	DOE 1/2 to D 10/2	92				
Z252	DOE 1/2 to D 11/2	42				
Z253	DOE 1/2 to D 12/2	95				
Z254	DOE 1/2 to D 13/2	43				
Z255	DOE 1/2 to D 14/2	94				
Z256	DOE 1/2 to D 15/2	44	4	50	nS	22
Low level to high impedance (tPLZ)						
Z257	-DOE 2/1 to D 0/1	57	6	61	nS	23
Z258	-DOE 2/1 to D 1/1	8				
Z259	-DOE 2/1 to D 2/1	58				
Z260	-DOE 2/1 to D 3/1	9				
Z261	-DOE 2/1 to D 4/1	59				
Z262	-DOE 2/1 to D 5/1	60				
Z263	-DOE 2/1 to D 6/1	11	6	61	nS	23

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TABLE XXVI. Disable time test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Fig
Z264	-DOE 2/1 to D 7/1	61	6	61	nS	23
Z265	-DOE 2/1 to D 8/1	73				
Z266	-DOE 2/1 to D 9/1	22				
Z267	-DOE 2/1 to D 10/1	71				
Z268	-DOE 2/1 to D 11/1	21				
Z269	-DOE 2/1 to D 12/1	23				
Z270	-DOE 2/1 to D 13/1	24				
Z271	-DOE 2/1 to D 14/1	74				
Z272	-DOE 2/1 to D 15/1	75				
Z273	-DOE 2/2 to D 0/2	30				
Z274	-DOE 2/2 to D 1/2	78				
Z275	-DOE 2/2 to D 2/2	28				
Z276	-DOE 2/2 to D 3/2	77				
Z277	-DOE 2/2 to D 4/2	27				
Z278	-DOE 2/2 to D 5/2	26				
Z279	-DOE 2/2 to D 6/2	76				
Z280	-DOE 2/2 to D 7/2	29				
Z281	-DOE 2/2 to D 8/2	91				
Z282	-DOE 2/2 to D 9/2	41				
Z283	-DOE 2/2 to D 10/2	92	6	61	nS	23

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TABLE XXVI. Disable time test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
Z284	-DOE 2/2 to D 11/2	42	6	61	nS	23
Z285	-DOE 2/2 to D 12/2	95				
Z286	-DOE 2/2 to D 13/2	43				
Z287	-DOE 2/2 to D 14/2	94				
Z288	-DOE 2/2 to D 15/2	44	6	61	nS	23
High level to high impedance (tPHZ)						
Z289	-DOE 2/1 to D 0/1	57	6	66	nS	23
Z290	-DOE 2/1 to D 1/1	8				
Z291	-DOE 2/1 to D 2/1	58				
Z292	-DOE 2/1 to D 3/1	9				
Z293	-DOE 2/1 to D 4/1	59				
Z294	-DOE 2/1 to D 5/1	60				
Z295	-DOE 2/1 to D 6/1	11				
Z296	-DOE 2/1 to D 7/1	61				
Z297	-DOE 2/1 to D 8/1	73				
Z298	-DOE 2/1 to D 9/1	22				
Z299	-DOE 2/1 to D 10/1	71				
Z300	-DOE 2/1 to D 11/1	21				
Z301	-DOE 2/1 to D 12/1	23				
Z302	-DOE 2/1 to D 13/1	24	6	66	nS	23

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TABLE XXVI. Disable time test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Fig
Z303	-DOE 2/1 to D 14/1	74	6	66	nS	23
Z304	-DOE 2/1 to D 15/1	75				
Z305	-DOE 2/2 to D 0/2	30				
Z306	-DOE 2/2 to D 1/2	78				
Z307	-DOE 2/2 to D 2/2	28				
Z308	-DOE 2/2 to D 3/2	77				
Z309	-DOE 2/2 to D 4/2	27				
Z310	-DOE 2/2 to D 5/2	26				
Z311	-DOE 2/2 to D 6/2	76				
Z312	-DOE 2/2 to D 7/2	29				
Z313	-DOE 2/2 to D 8/2	91				
Z314	-DOE 2/2 to D 9/2	41				
Z315	-DOE 2/2 to D 10/2	92				
Z316	-DOE 2/2 to D 11/2	42				
Z317	-DOE 2/2 to D 12/2	95				
Z318	-DOE 2/2 to D 13/2	43				
Z319	-DOE 2/2 to D 14/2	94				
Z320	-DOE 2/2 to D 15/2	44	6	66	nS	23
Low level to high impedance (tPLZ)						
Z321	-DOE 3/1 to D 0/1	57	6	61	nS	23
Z322	-DOE 3/1 to D 1/1	8	6	61	nS	23

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TABLE XXVI. Disable time test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
Z323	-DOE 3/1 to D 2/1	58	6	61	nS	23
Z324	-DOE 3/1 to D 3/1	9				
Z325	-DOE 3/1 to D 4/1	59				
Z326	-DOE 3/1 to D 5/1	60				
Z327	-DOE 3/1 to D 6/1	11				
Z328	-DOE 3/1 to D 7/1	61				
Z329	-DOE 3/1 to D 8/1	73				
Z330	-DOE 3/1 to D 9/1	22				
Z331	-DOE 3/1 to D 10/1	71				
Z332	-DOE 3/1 to D 11/1	21				
Z333	-DOE 3/1 to D 12/1	23				
Z334	-DOE 3/1 to D 13/1	24				
Z335	-DOE 3/1 to D 14/1	74				
Z336	-DOE 3/1 to D 15/1	75				
Z337	-DOE 3/2 to D 0/2	30				
Z338	-DOE 3/2 to D 1/2	78				
Z339	-DOE 3/2 to D 2/2	28				
Z340	-DOE 3/2 to D 3/2	77				
Z341	-DOE 3/2 to D 4/2	27				
Z342	-DOE 3/2 to D 5/2	26				
Z343	-DOE 3/2 to D 6/2	76	6	61	nS	23

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TABLE XXVI. Disable time test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
Z344	-DOE 3/2 to D 7/2	29	6	61	nS	23
Z345	-DOE 3/2 to D 8/2	91				
Z346	-DOE 3/2 to D 9/2	41				
Z347	-DOE 3/2 to D 10/2	92				
Z348	-DOE 3/2 to D 11/2	42				
Z349	-DOE 3/2 to D 12/2	95				
Z350	-DOE 3/2 to D 13/2	43				
Z351	-DOE 3/2 to D 14/2	94				
Z352	-DOE 3/2 to D 15/2	44	6	61	nS	23
High level to high impedance (tPHZ)						
Z353	-DOE 3/1 to D 0/1	57	6	66	nS	23
Z354	-DOE 3/1 to D 1/1	8				
Z355	-DOE 3/1 to D 2/1	58				
Z356	-DOE 3/1 to D 3/1	9				
Z357	-DOE 3/1 to D 4/1	59				
Z358	-DOE 3/1 to D 5/1	60				
Z359	-DOE 3/1 to D 6/1	11				
Z360	-DOE 3/1 to D 7/1	61				
Z361	-DOE 3/1 to D 8/1	73				
Z362	-DOE 3/1 to D 9/1	22	6	66	nS	23

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TABLE XXVI. Disable time test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
Z363	-DOE 3/1 to D 10/1	71	6	66	nS	23
Z364	-DOE 3/1 to D 11/1	21				
Z365	-DOE 3/1 to D 12/1	23				
Z366	-DOE 3/1 to D 13/1	24				
Z367	-DOE 3/1 to D 14/1	74				
Z368	-DOE 3/1 to D 15/1	75				
Z369	-DOE 3/2 to D 0/2	30				
Z370	-DOE 3/2 to D 1/2	78				
Z371	-DOE 3/2 to D 2/2	28				
Z372	-DOE 3/2 to D 3/2	77				
Z373	-DOE 3/2 to D 4/2	27				
Z374	-DOE 3/2 to D 5/2	26				
Z375	-DOE 3/2 to D 6/2	76				
Z376	-DOE 3/2 to D 7/2	29				
Z377	-DOE 3/2 to D 8/2	91				
Z378	-DOE 3/2 to D 9/2	41				
Z379	-DOE 3/2 to D 10/2	92				
Z380	-DOE 3/2 to D 11/2	42				
Z381	-DOE 3/2 to D 12/2	95				
Z382	-DOE 3/2 to D 13/2	43	6	66	nS	23

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TABLE XXVI. Disable time test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Fig
Z383	-DOE 3/2 to D 14/2	94	6	66	nS	23
Z384	-DOE 3/2 to D 15/2	44	6	66	nS	23

NOTES:

1. Disable time tolerance table.

Condition	Value	Accuracy	Test condition
Power supply	5.0 V	± 50 mV	Z1-Z384
Power supply current limit	0.878 A	NA	Z1-Z384
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	Z1-Z384
low, (VL)	0.0 V	± 100 mV	Z1-Z384
Output time measurement	NA	± 2.1 nS	Z1-Z32, Z193-Z224
	NA	± 2.3 nS	Z33-Z64, Z225-Z256
	NA	± 2.8 nS	Z65-Z96, Z129-Z160, Z257-Z288, Z321-Z352
	NA	± 3.0 nS	Z97-Z128, Z161-Z192, Z289-Z320, Z353-Z384

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line. Module inputs shall not violate the minimum input timing parameters referenced on figure 9.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

3. Module inputs shall meet the timing requirements shown on figures 22 and 23.

4. Output pins 8, 9, 11, 12, 13, 14, 16, 17, 18, 21, 22, 23, 24, 26, 27, 28, 29, 30, 32, 34, 35, 36, 37, 38, 39, 41, 42, 43, 44, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 66, 67, 68, 71, 73, 74, 75, 76, 77, 78, 81, 82, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, and 95 shall be loaded as shown on figure 8.

5. Abbreviations used are defined in 6.1.

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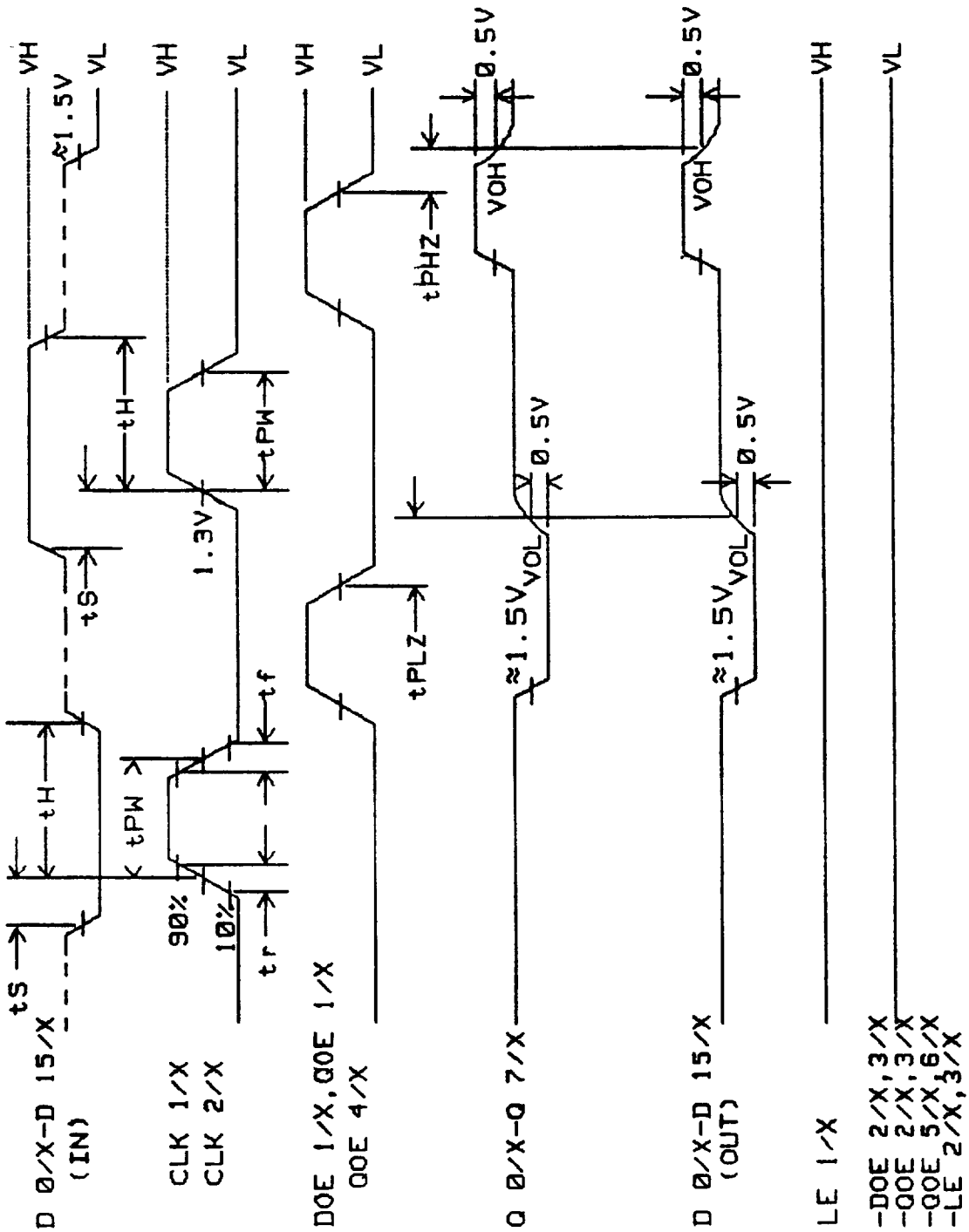


FIGURE 22. Disable time test waveforms for table XXVI.

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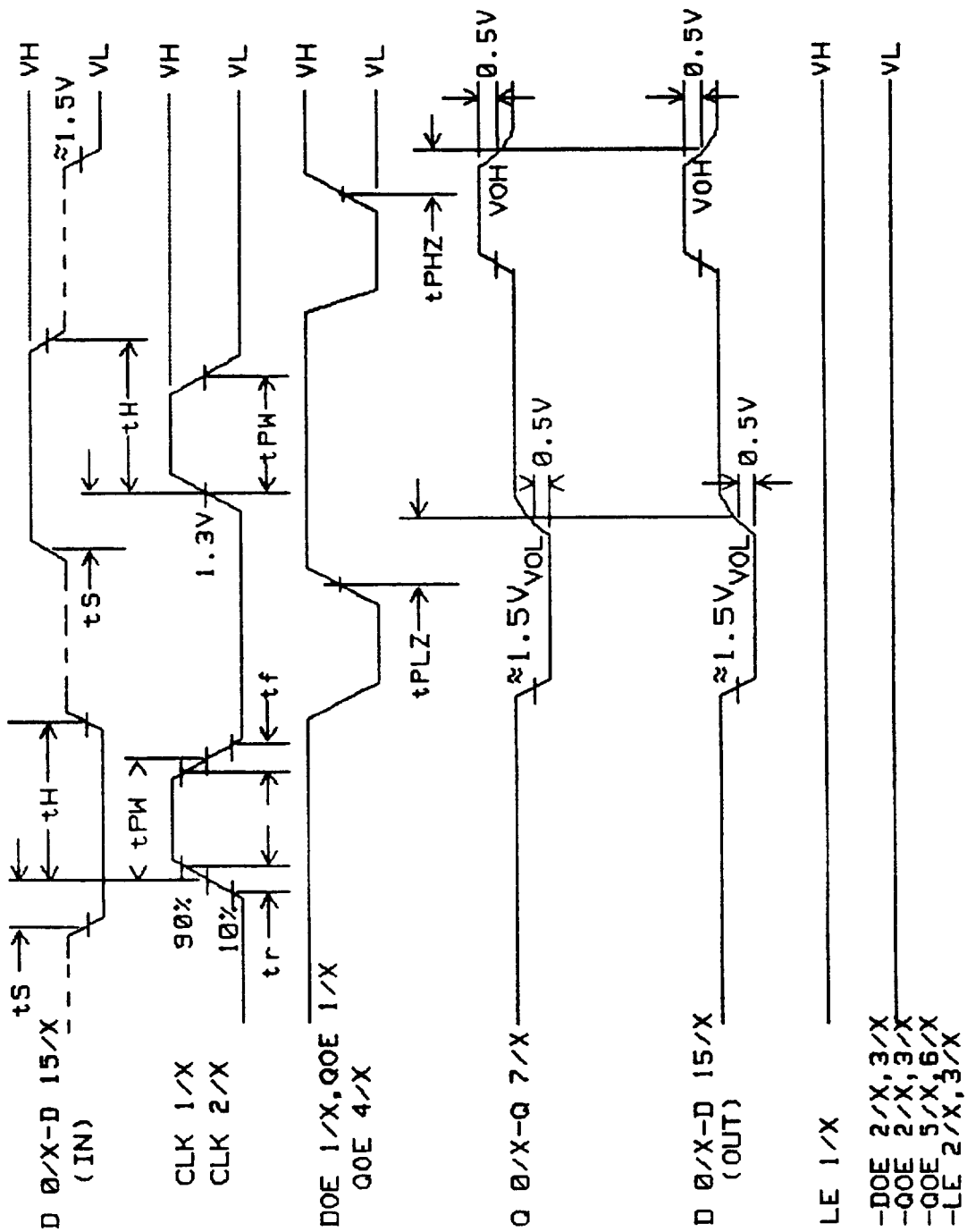


FIGURE 23. Disable time test waveforms for table XXVI.

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NOTES:

1. Input parameters for figures 22 and 23:

   tS = 20 nS min.
   tH = 5 nS min.
   tPW = 20 nS min.
   tr = 15 nS max.
   tf = 6 nS max.

2. Pattern voltage shall be as specified in the test tolerance table.

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TABLE XXVII. Noise margin test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
AA1	Q 0/1	63	0	900	mV	1-4	24,25
AA2	Q 1/1	13					
AA3	Q 2/1	62					
AA4	Q 3/1	12					
AA5	Q 4/1	7					
AA6	Q 5/1	56					
AA7	Q 6/1	6					
AA8	Q 7/1	55					
AA9	Q 8/1	17					
AA10	Q 9/1	18					
AA11	Q 10/1	68					
AA12	Q 11/1	67					
AA13	Q 12/1	64					
AA14	Q 13/1	14					
AA15	Q 14/1	16					
AA16	Q 15/1	66					
AA17	Q 0/2	87					
AA18	Q 1/2	37					
AA19	Q 2/2	86					
AA20	Q 3/2	35	0	900	mV	1-4	24,25

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TABLE XXVII. Noise margin test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
AA21	Q 4/2	82	0	900	mV	1-4	24,25
AA22	Q 5/2	32					
AA23	Q 6/2	81					
AA24	Q 7/2	34					
AA25	Q 8/2	39					
AA26	Q 9/2	89					
AA27	Q 10/2	93					
AA28	Q 11/2	90					
AA29	Q 12/2	38					
AA30	Q 13/2	88					
AA31	Q 14/2	36					
AA32	Q 15/2	85	0	900	mV	1-4	24,25

NOTES:

1. Noise margin test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 275 mV	AA1-AA32
Power supply current limit	0.878 A	NA	AA1-AA32

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 AA1-AA32 AA1-AA32
Noise margin measurement	NA	± 45 mV	AA1-AA32

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line. Module inputs shall not violate the minimum input timing parameters referenced on figure 9.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

(d) For each output under test in table XXVII, connect probe 'B' and visually inspect the output for the most positive spike as indicated on figure 24.

(e) The maximum probe capacitance is 10 pF.

3. The test circuit shall be as indicated on figure 25. All outputs shall be loaded as specified on figure 8.

4. Abbreviations used are defined in 6.1.

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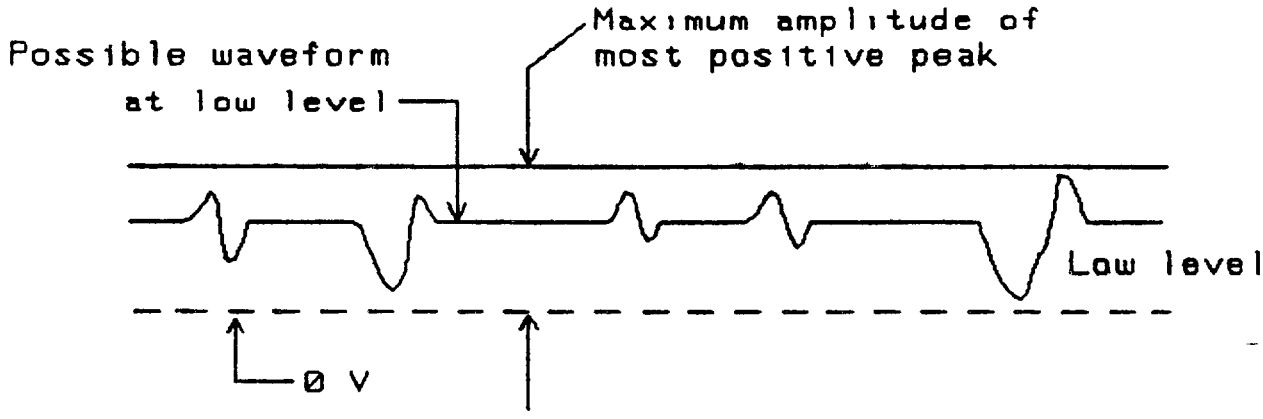


FIGURE 24. Noise margin test waveforms for table XXVII.

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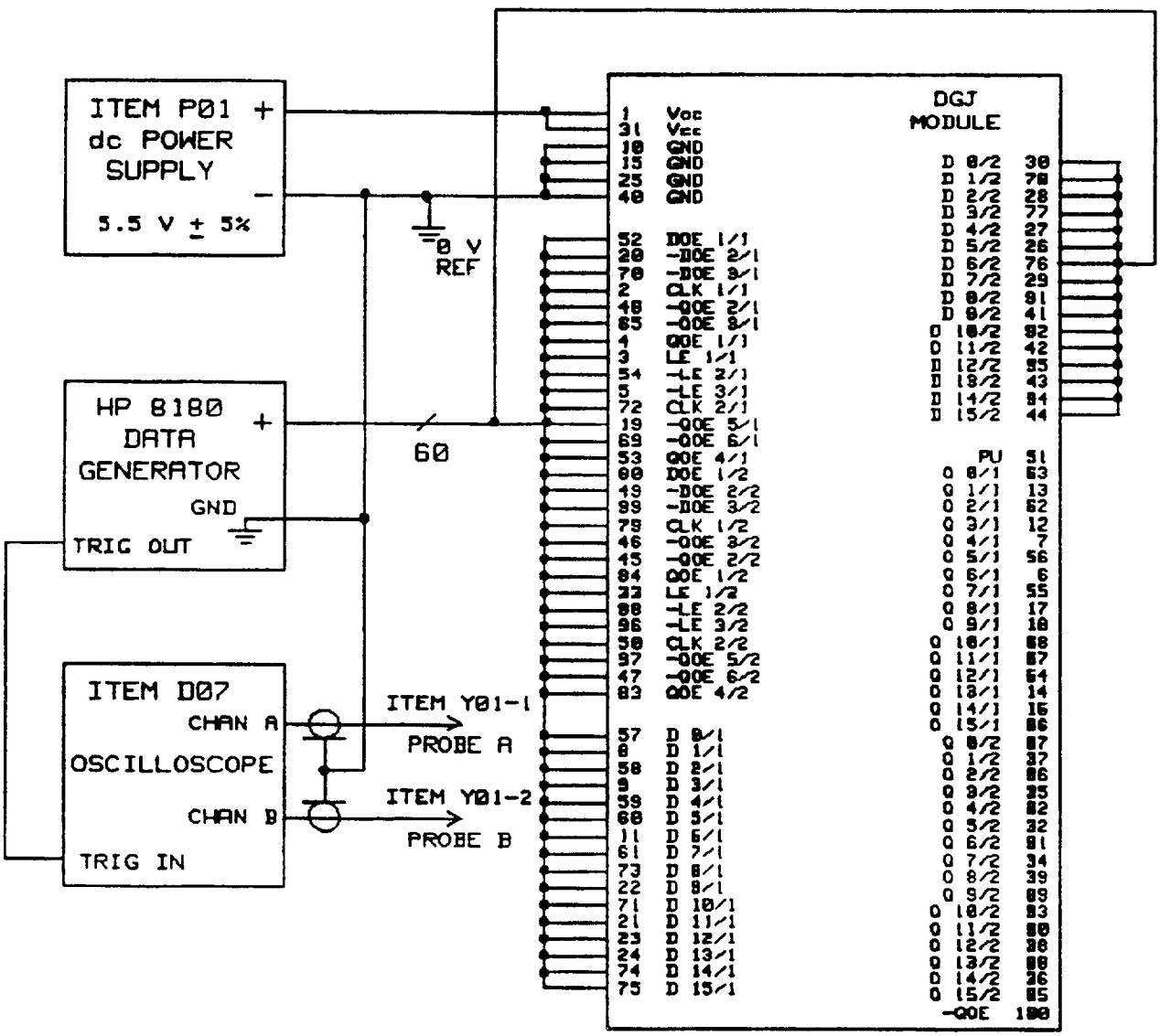


FIGURE 25. Noise margin test circuit for XXVII.

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